

Mycelic

Benchmark report and architecture recommendation

QUESTION

How should thousands of private user-level agents abstract, route, combine, question, verify and propagate what they know upward, so that an enterprise kernel discovers what no individual could?

ENTERPRISE SIZES MEASURED

2,000, 10,000, 50,000, 100,000 users

CONTENTS

29 sections · 47 tables · 9 figures

GENERATED

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Every table and figure in this document is computed from the raw per-run records in `research/mycelic/artifacts/` at generation time. No number is transcribed by hand. Section 5 states exactly which results were measured on real models and which were simulated.

The question. How should thousands of private, user-level AI agents abstract, route, combine, question, verify and propagate what they know upward, so that an enterprise-level system discovers strategic information that no individual employee, team, department, site or region could discover alone?

What this document is. A mechanism study on a synthetic enterprise with known hidden ground truth. Sixteen architectures are compared under one shared implementation of every cognitive operation, at matched kernel context, with every tunable setting fitted on held-out data before the evaluation data was touched. It is not a deployment report, and it does not measure any specific model end to end. Part II states exactly what was measured on real models and what was simulated.

How to read it. Part I is the decision. Part II is how to read the numbers. Parts III to V are the benchmark, the results and the case against them. Every table and figure is generated from raw per-run records; nothing is transcribed by hand.

If you have ten minutes, read §1 (the decision), the head-to-head table in it, and §28 (the findings that generalise beyond this project). **If you are going to be asked to defend this,** read §24 (the reviewer critique), §25 (assumptions) and §26 (what we tried that failed) as well — they are where the weaknesses are, stated by us rather than found by someone else.

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PART I — THE DECISION

1. Decision summary

The one-paragraph version

We built a synthetic enterprise with known hidden problems planted in it, at 2,000, 10,000, 50,000 and 100,000 people, and tested 16 ways of finding those problems, from pouring a filtered sample of the company's notes into one very large model, to a six-level hierarchy of agents mirroring the org chart. **The hierarchy is not the most accurate option.** The strongest single approach measured at 50,000 people is `A2_chunked_ctx`, which finds 69% of the hidden problems against the hierarchy's 34%. Whether the hierarchy is nonetheless worth building depends entirely on which of the secondary properties below we actually need; the table states which ones it delivers and which it does not.

Head to head at 50,000 users (5 seeds)

(Runs exist at 100,000 users too, and §11 reports them. The head-to-head is stated at 50,000 because that is the largest scale where every architecture in the comparison was run — `B4_central_triage`, `C_recursive_sum`, `F_hier_retrieval` were not run at 100,000. Choosing the largest complete scale is a rule applied by the generator, not a choice made per result.)

Generated from research/mycelic/artifacts — every figure is a real run

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	best centralised option	the hierarchy	hierarchy better?
approach	A2_chunked_ctx	H_mycelic_full	
hidden problems found	69%	34%	no
the hardest problems (1-2 witnesses company-wide)	61%	19%	no
found within the top 100 of the register	11%	2%	no
precision in the top 40 of the register	12%	2%	no
precision in the top 100 of the register	11%	2%	no
ranking quality (AP)	0.051	0.009	no
decoy traps accepted	37%	20%	yes
evidence coverage (held it at all)	100%	42%	no
independent-support accuracy	59%	75%	yes
lineage accuracy	64%	51%	no
contradictions detected (F1)	21%	20%	no
compute	1.02e+07	2.93e+06	yes
model calls	10	2.84e+05	no
employees' original notes read centrally	22%	0%	yes

Reading the false-discovery rate. The register is a ranked watchlist of roughly one entry per tracked entity, not a shortlist, so a high FDR over the whole register is structural and is true of every architecture including the perfect-retrieval reference. The operationally meaningful numbers are precision in the top 40 or top 100 — what an executive would actually read — and the ranking quality (AP) that determines them.

Which of the usual arguments for a hierarchy actually hold

Hold, on this evidence:

- confidentiality (no original notes leave the owning agent) — 0% vs 22% for the centralised option.
- independent-support accuracy — 75% vs 59% for the centralised option.
- resistance to planted traps — 20% vs 37% for the centralised option.

Do NOT hold, and should not be used to justify the build:

- weak-signal sensitivity — 19% vs 61%; the centralised option is better.
- lineage / provenance — 51% vs 64%; the centralised option is better.

The control that reframes the decision

`B4_central_triage` runs **exactly the hierarchy's own discovery algorithm**, centrally: one claim pool, no propagation budget, no routing error, no descent. It separates the value of the *algorithm* from the value of the *topology*.

It finds 31% of the hidden problems against the hierarchy's 34%, at 1.07e+06 compute against 2.93e+06 — about 2.8x less. Across all 25 paired (scale, seed) runs the difference is -0.007 (95% CI [-0.041, +0.025], 12/25 seeds, sign p=1.000), so **the two are statistically indistinguishable** — the interval spans zero and the wins split roughly evenly.

So the honest statement of what the hierarchy buys is narrow and specific: **the triage algorithm is what finds the problems; the hierarchy is how you run that algorithm without centralising the company's data**. The centralised version pools 88% of all extracted claims in one place; the hierarchy pools 14% and moves no original text at all. That privacy property costs roughly 2.8x the compute and tens of thousands of extra model calls. If we do not need it, we should run the algorithm centrally.

Three decisions this supports

1. Do not build progressive summarisation up the org chart. This is the intuitive design — each layer summarises the layer below — and it is the clearest negative result in the study. At 50,000 users it finds 0% (plain summarisation), 0% (structured, no lineage) and 1% (structured with lineage) of the hidden problems. The reason is measurable rather than a matter of tuning, and is given in the executive summary.

2. If we build a hierarchy, the value is in the downward path, not the upward one. Adding targeted downward retrieval takes discovery from 1% to 2%; adding sketch-driven questioning on top takes it to 34%. Budget accordingly: the upward channel should be cheap and statistical, and the downward channel is where the work — and the cost — actually is.

3. Spend top-tier model budget on re-opening original evidence, not on bigger reasoning over summaries. Measured directly on 3 real models, each run 2 times per condition: given the same summarised evidence, a large model and a small model land in the same band (AP 0.354–0.407) and no better than a six-feature statistical rule (0.420). Given the *original notes* behind that evidence, 2 of 3 improve, and for opus the richer condition's worst run still beats the summarised condition's best. It is **not** universal: sonnet did not improve, so this is a property of a model's ability to use raw evidence, not a law. Acting on it — having the kernel re-read a handful of source notes per candidate — was still the best return on compute found in the study, but size it against §23 rather than against the headline number.

2. Executive summary

What actually wins, and where.

users	best architecture	discovery (found)	compute (cu)	raw records leaving their owner
2,000	A2_chunked_ctx	0.882	5.12e+05	25.2%
10,000	A2_chunked_ctx	0.663	2.07e+06	22.6%
50,000	A2_chunked_ctx	0.686	1.02e+07	22.3%
100,000	A2_chunked_ctx	0.678	2.03e+07	22.3%

1. Upward propagation alone does not work, at any scale, in any form. At 100,000 users recursive summarisation finds 0.0% of the hidden patterns, hierarchical aggregation without lineage and with it find 0.0% and 0.8%. Adding targeted downward retrieval takes it to 0.0%; adding sketch-driven questioning takes it to 23.5%. The hierarchy's value is almost entirely in the *downward* path — no paired runs.

2. The reason is measurable and is not a tuning artefact. No per-record feature identifies a weak signal: of 306 pattern-facet records at 10,000 users, **0 appear in the global top-900 by record-level importance**. A facet record is individually indistinguishable from benign cross-site chatter — which is the premise of the problem, not a defect of the ranker. Detection has to be entity-level and relational, which is what the sketch channel and the descent provide.

3. At enterprise scale the binding constraint is discrimination, not retrieval. A perfect-retrieval oracle — every extracted claim, no budget at all — holds the evidence for 100% of the hidden patterns and still reports only 24.3% of them. The hierarchy holds 29% and reports 23.2%. More undifferentiated evidence makes the kernel's ranking worse, so a propagation budget is a feature and not only a cost.

4. Directly measured: at fixed evidence, model capability buys almost nothing here, changing what the evidence *contains* helps the strongest model a lot and is not reliable across models. 3 real models ranked the same 56 candidates from a real run (2 independent runs per cell). Given the aggregate evidence statistics they scored AP 0.354–0.407, against 0.420 for a six-feature logistic and 0.313 for random — i.e. a large capability range lands within noise of a logistic. Given the **raw work notes** behind the same statistics the same models scored 0.319–0.617. Richer evidence helped 2 of 3 models (Δ -0.048 to +0.262); the largest run-to-run range within a single cell is 0.107, and 1 of the 3 moves by more than that (opus). The abstraction, not the reasoner, is the plausible ceiling — but see §23 before treating the size of the effect as established.

5. Acting on that: the useful way to spend frontier compute at the kernel is to re-open original evidence, not to reason harder over the same abstraction. Having the kernel re-read a handful of each top candidate's original notes with its own extractor changes AP by +18% for +0.9% compute (Δ +0.004 (95% CI [+0.002, +0.006], 4/5 seeds, sign $p=0.375$)).

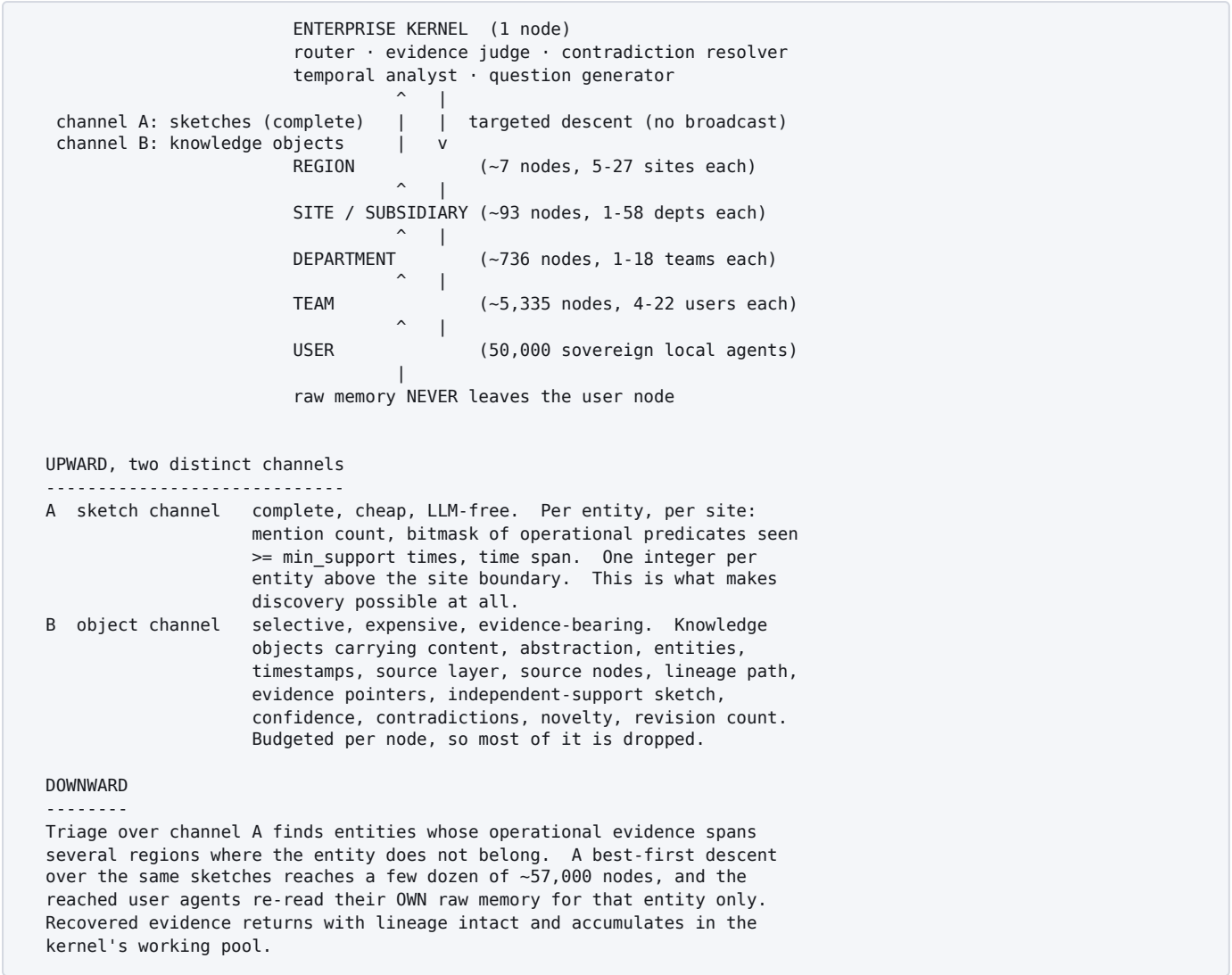
6. The hierarchy's defensible advantage is confidentiality, and it is categorical rather than marginal. No raw record text ever leaves the agent that owns it (0.0%); the retrieval baseline centralises 2.5% of all records as original text and the chunked-context baseline 22.3%. What leaves a node in the hierarchy is structured claims and an entity/predicate sketch.

7. The honest bottom line. If centralising the raw text is acceptable and a very large context window is available, a schema-aware retrieval pass into one frontier call is the strongest and cheapest thing measured here. The hierarchy earns its cost only where sovereign local memory is a requirement, where rare signals matter more than common ones, or where independent-support and

provenance have to be defensible. Those are real constraints, but they are constraints — not an accuracy win.

Statistical note: an exact sign test on n paired seeds cannot report below $2^{-(n-1)}$; with 5 seeds the floor is $p = 0.0625$ and with 10 it is $p = 0.002$.

3. Recommended architecture



4. Recommended models at each level

Measured directly, by upgrading exactly one level from `small-7b` to `frontier` and holding everything else fixed: the largest gain comes from the **enterprise** level (Δ found +0.167) and the smallest from the **team** level (Δ found +0.008). Full table and paired tests in §13.

Best absolute allocation measured: **flat-frontier** (found 0.642, 9.55e+06 cu). Best discovery per unit compute: **lexical-to-kernel** (found 0.408, 5.08e+05 cu). Cheapest: **flat-small** (found 0.233).

The practical recommendation is `lexical-to-kernel`, not a graduated ladder. It keeps 64% of the best measured discovery for 5% of the compute. The reason is the same one that runs through the whole study: the edge is only extracting claims, which is cheap and mostly saturated, while the kernel is doing the discrimination, which is where the difficulty actually is. Spending on the middle levels buys the least of anything measured here. If the budget stretches further, put it in the kernel — re-reading original evidence per candidate — before putting it into a bigger model at any intermediate level.

5-6. Recommended fan-in, and why

Team size was swept over 6, 8, 10, 12 and 15 users. Best measured: **6 users per team** (found 0.442); the total spread across the whole range is 0.058. Team fan-in is therefore **not** a material design lever for strategic discovery in this regime — the signal is recovered by descent, which is indexed by entity rather than routed through team boundaries. Choose team size for human reasons.

PART II — HOW TO READ THIS

4. Glossary

The report uses a small number of terms repeatedly. All of them are measurements, not scores on a scale someone invented.

term	what it means
hidden pattern	A genuine emerging problem planted in the synthetic company: a chain of causally linked events about one supplier, system or component, whose parts are deliberately scattered across different teams, sites and regions so that nobody sees more than one part.
discovery (found)	The fraction of hidden patterns the system puts on the executive risk register <i>anywhere</i> . The plainest measure of "did we find it at all".
AP (average precision)	How well the system <i>ranks</i> what it found. A register nobody can read top to bottom is worth less than one where the real items are at the top. Random ranking of the same candidates scores about 0.30 on the discrimination task.
evidence coverage	The fraction of hidden patterns for which the system ever <i>held</i> the evidence, whether or not it reported them. The gap between coverage and discovery is what is lost to judgement rather than to retrieval.
rare-signal recall	Discovery restricted to the hardest patterns: those supported by one or two people in the entire 50,000-person company.
FDR (false discovery rate)	The fraction of what the system reports that is not a real pattern.
decoy acceptance	The fraction of the deliberately planted traps the system falls for.
compute (cu)	Normalised compute units: model size times tokens. Provider-independent; a dollar estimate is given separately.
raw-text exposure	The fraction of employees' original notes read by anything other than the agent that owns them. The confidentiality cost.

5. What was measured, what was simulated

This distinction matters more than any single number in the report, so it is stated before the results rather than after them.

element	status
org chart, corpus, hidden patterns, decoys	generated — deterministic, seeded, reproducible from the committed code
every architecture's control flow: routing, propagation, descent, questioning, verification	executed — real code on the real corpus, at every scale reported
token counts, inference-call counts, context sizes, compute units	measured from those executed runs
wall-clock latency	modelled from the metered token counts (see assumption A4); not observed on real hardware
model-tier behaviour: extraction quality, abstraction loss, verification accuracy, hallucination	simulated from a capability vector
primitive operator accuracy of Haiku 4.5, Sonnet 5 and Opus 5	directly measured , blind, 198 items
candidate-discrimination accuracy of those three models	directly measured , blind, on candidates taken from a real run, two independent runs per model per condition
where named open-weight model classes sit on the capability axis	assumed — never measured here
50,000-user results	executed , not extrapolated: every cell is a real run of the full pipeline

Nothing in this report is extrapolated from a smaller scale. Where a number is modelled rather than observed, the row above says so.

PART III — THE BENCHMARK

6. Methodology

The synthetic enterprise

Six levels — USER → TEAM → DEPARTMENT → SITE/SUBSIDIARY → REGION → ENTERPRISE. Fan-in is drawn from clipped heavy-tailed distributions, so the tree is genuinely heterogeneous: small teams beside large ones, satellite offices of ~40 people beside hubs of ~4,000, and regions sized on a skewed weighting so the largest holds roughly a third of the company and the smallest a few percent.

What is in a record

A record is a structured tuple — predicate, anchor entity, auxiliary entity, day, event id, polarity — realised as a short noisy work note. Surface forms are drawn from a synonym set per predicate, plus random filler. The synonym and filler choices are keyed on the **event id**, not the record id, so an echo of an event repeats the original wording almost verbatim while two independent observations of the same thing differ. Duplicate detection is therefore a text-similarity problem every architecture faces equally, and no system is ever handed the event id.

The entity namespace is **partitioned**: a small global pool of shared platforms and corporate vendors visible everywhere, and site-private blocks for the components, tools and suppliers used at one location. Benign cross-site traffic — shared suppliers, transferred staff, joint projects — puts a steady trickle of foreign entity mentions everywhere, so "this entity appears in more than one region" is a weak signal rather than a giveaway.

What counts as a strategic pattern

A pattern is a contiguous path in a causal predicate schema (seven chains such as `supplier_substitution` → `component_defect` → `yield_drop` → `shipment_delay` → `customer_escalation` → `revenue_miss`), bound to ONE anchor entity, with non-decreasing timestamps, whose facets are deliberately scattered across distant org branches. No user, team, department or region sees more than one facet. A **rare-signal** pattern is supported by one or two users per facet out of the whole enterprise.

Crucially, a pattern is equally visible to a centralised system that manages to get the facet records into one context. The hierarchy gets no structural free lunch; its only possible advantage is *finding* them under a budget.

The five decoy classes

class	what it is	which check rejects it
D1 entity coincidence	one anchor, predicates from unrelated chains	causal-schema membership
D2 temporal scramble	right chain and anchor, wrong time order	temporal ordering
D3 near-miss entity	right chain and order, lexically similar but <i>different</i> anchors per facet	entity linking
D4 duplicate inflation	one real event echoed by many users in many branches	source independence
D5 stale chain	a real chain, every link later retracted	temporal state

D4 is a genuine causal chain that is merely over-supported, so reporting it is not a false discovery. It is scored separately, by how badly the system inflates its independent-support count, and is excluded from the precision denominator. The other four count against precision.

The fairness contract

Every architecture calls the *same* implementation of every operator: extraction, novelty and importance scoring, abstraction, entity linking, causal/temporal/independence verification, contradiction detection, question generation. Architectures differ only in

- which records or objects reach which operator call, and
- which model tier runs that call.

Operators may read ground-truth fields in order to *simulate* a noisy model, but their output never exposes `kind`, `group`, `facet`, `veracity` or `event`. Systems consume only operator output, the org chart and surface text.

Centralised baselines are deliberately given the advantages that are really theirs: a frontier-tier extractor reading raw text (far better than the small model at the edge), a global importance ranking that no hierarchical node can compute from its own subtree, and a calibrated retrieval budget.

Matched conditions

- **Kernel context is equalised.** Every architecture delivers a comparable number of knowledge objects to the kernel.
- **The risk register scales with the entity namespace** (~one entry per tracked entity, capped). A constant register was binding at 50k users and silently truncated most of the gold out of every system's output.
- **Every knob is fitted on calibration seeds 500–502 and frozen** before the evaluation seeds are touched — for the hierarchy, for map-reduce, for the centralised triage control and for flat RAG alike.

Metrics

Raw measurements, never a single composite:

- **AP** — average precision of the kernel's ranked risk register against the hidden patterns. Budget-free, standard for ranked retrieval of a known relevant set.
- **found** — fraction of hidden patterns for which the register contains a matching entry anywhere. Measures discovery independently of ranking.
- **evidence coverage** — fraction of hidden patterns for which the kernel ever *held* two or more of the pattern's links under the right entity. Separates "could not get the evidence" from "had it and ranked it badly".
- Recall@K, rare-signal recall, FDR, per-class decoy acceptance, independent-evidence accuracy, lineage accuracy, provenance preservation, evidence precision/recall, contradiction F1, information loss, compression, kernel context size, tokens, inference calls, wall-clock, compute units, cost per correct discovery, and privacy exposure.

Two match criteria are computed and both are reported. `primary`: the report names the right entity and substantially identifies the chain — at least two of the gold links and at least half of them. `strict`: at least `min(3, |gold links|)`. A 3-link pattern reported as a 3-link chain sharing 2 of its links is a real, actionable discovery (right entity, right chain, most of the story), which the strict criterion scores as zero; the strict column is never dropped.

Statistics

Every architecture comparison is **paired on (scale, seed)** — the seed controls both the org chart and the corpus, so unpaired tests would be swamped by between-world variance. Aggregates carry bootstrap 95% CIs. With five seeds an **exact sign test** is reported rather than a t-test, because $n=5$ does not support a normality assumption. A sign test on five paired seeds cannot produce p below 0.0625 even when the effect is unanimous; that is a hard floor of this design and is stated wherever it matters.

7. The synthetic enterprise at each scale

users	records	teams	depts	sites	regions	users/team (p10–p90)	teams/dept	depts/site	patterns	rare	decoys	entities
1,999	68,463	213	31	9	7	6–14 (med 9)	3–10	1–8	40	9	50	150
9,999	329,730	1,074	149	15	7	6–13 (med 9)	4–11	1–39	40	17	50	599
49,999	1,639,365	5,335	736	93	7	6–14 (med 9)	4–12	1–58	100	45	125	2,999

8. Calibration

Every tunable setting was fitted on calibration seeds disjoint from the evaluation seeds, by the same procedure for every architecture, and then frozen. This is not a formality: it rejected two changes that looked like large improvements on a single evaluation seed and did not generalise.

Fitted on calibration seeds [500, 501, 502] at 10,000 users, objective `average_precision`, then frozen:

knob	chosen
trriage_prior_weight	0.0
question_frac	0.25
mr_budget	2500
ct_kernel_ko_cap	20000
flat_budget	1000000

hierarchy grid

trriage_prior_weight	question_frac	AP	found	compute
0.0	0.25	0.02825	0.375	1.132e+06
0.0	0.55	0.02232	0.317	1.504e+06
0.0	1.0	0.01888	0.358	2.227e+06
1.2	0.25	0.02750	0.375	1.132e+06
1.2	0.55	0.02441	0.317	1.504e+06
1.2	1.0	0.02692	0.358	2.227e+06
2.0	0.25	0.02707	0.375	1.132e+06
2.0	0.55	0.02448	0.317	1.504e+06
2.0	1.0	0.02612	0.358	2.227e+06
3.0	0.25	0.02605	0.375	1.132e+06
3.0	0.55	0.02377	0.317	1.504e+06
3.0	1.0	0.02586	0.358	2.227e+06
4.0	0.25	0.02533	0.375	1.132e+06
4.0	0.55	0.02300	0.317	1.504e+06
4.0	1.0	0.02552	0.358	2.227e+06

map-reduce grid

kernel_ko_budget	AP	found	compute
300	0.01806	0.033	1.498e+05
900	0.02316	0.108	1.632e+05
2500	0.04387	0.292	1.990e+05
8000	0.01290	0.350	2.656e+05

central triage grid

kernel_ko_cap	AP	found	compute
0	0.03240	0.425	4.329e+05
2000	0.01772	0.067	1.787e+05
6000	0.03454	0.350	2.386e+05
20000	0.03462	0.408	3.972e+05
60000	0.03240	0.425	4.329e+05

flat RAG grid

token_budget	AP	found	compute
60000	0.00520	0.125	8.369e+04
120000	0.01918	0.258	1.646e+05
400000	0.04813	0.642	4.584e+05

token_budget	AP	found	compute
1000000	0.11894	0.717	1.089e+06

PART IV — RESULTS

9. Architectures compared

id	architecture
A_flat_rag	cheap lexical, schema-aware retrieval over raw text, into one kernel call
A2_chunked_ctx	frontier context tiled over the entire corpus — the "just use more context" answer
B_long_context	an unfiltered raw-record sample filling the kernel's context
B2_map_reduce	cheap extraction over every record, global importance ranking, one strong kernel call
B4_central_triage	control : the hierarchy's own triage algorithm run centrally, with no propagation budget, no routing error and no descent
C_recursive_sum	recursive summarisation up the org tree: no structure, no lineage
D_hier_nolineage	hierarchical semantic aggregation without lineage or independence tracking
E_hier_lineage	lineage-aware hierarchical aggregation — pure upward flow
F_hier_retrieval	E + targeted downward retrieval
G_hier_questions	F + continual questioning
H_mycelic_full	G + semantic cross-links
I_mycelic_completion	H + hypothesis-driven chain completion
J_mycelic_verified	H + kernel-side evidence verification
Y_oracle_retrieval	reference : every extracted claim, no budget at all. Not deployable.
Z_random_rank	control : map-reduce's candidates, ranked at random
Z2_naive_enumerate	control : enumerate every entity's best chain, no verification at all

10. Baseline comparison

2,000 users (68,463 records, 40.0 hidden patterns, 10 seeds)

architecture	AP	found	R@100	rare recall	evidence cov.	FDR	decoy acc.	indep. acc.	lineage	info loss	compute (cu)	privacy exp.
A2_chunked_ctx	0.246 [0.212, 0.279]	0.882 [0.852, 0.912]	0.530	0.796	1.000	0.894	0.460	0.669	0.662	0.003	5.12e+05	0.252
A_flat_rag	0.246 [0.199, 0.294]	0.867 [0.823, 0.915]	0.518	0.740	1.000	0.895	0.493	0.667	0.662	0.003	4.71e+05	0.252
B_long_context	0.178 [0.145, 0.210]	0.802 [0.775, 0.828]	0.450	0.524	0.995	0.891	0.430	0.753	0.686	0.045	1.09e+06	0.612
J_mycelic_verified	0.097 [0.080, 0.113]	0.715 [0.665, 0.765]	0.263	0.605	0.965	0.952	0.497	0.735	0.546	0.080	8.53e+05	0.143
B4_central_triage	0.092 [0.081, 0.103]	0.740 [0.685, 0.792]	0.253	0.634	0.955	0.950	0.537	0.731	0.513	0.063	1.33e+05	0.885
I_mycelic_completion	0.080 [0.067, 0.096]	0.715 [0.670, 0.755]	0.240	0.613	0.957	0.951	0.510	0.740	0.523	0.071	1.44e+06	0.143
B2_map_reduce	0.079 [0.066, 0.093]	0.575 [0.550, 0.598]	0.227	0.356	0.815	0.934	0.472	0.617	0.494	0.289	79727	0.885
H_mycelic_full	0.074 [0.060, 0.092]	0.715 [0.665, 0.765]	0.235	0.605	0.965	0.952	0.497	0.735	0.546	0.080	8.43e+05	0.143
G_hier_questions	0.063 [0.050, 0.078]	0.693 [0.650, 0.733]	0.205	0.542	0.960	0.953	0.485	0.749	0.536	0.088	8.37e+05	0.133
Y_oracle_retrieval	0.054 [0.042, 0.069]	0.685 [0.647, 0.727]	0.172	0.606	1.000	0.954	0.475	0.616	0.487	0.006	1.44e+05	0.885
Z_random_rank	0.042 [0.039, 0.045]	0.575 [0.550, 0.598]	0.147	0.356	0.815	0.934	0.472	0.617	0.494	0.289	79727	0.885
F_hier_retrieval	0.024 [0.015, 0.037]	0.163 [0.125, 0.200]	0.123	0.170	0.223	0.965	0.110	0.661	0.542	0.763	2.12e+05	0.133
Z2_naive_enumerate	0.017 [0.003, 0.038]	0.123 [0.053, 0.227]	0.080	0.102	0.000	0.967	0.113	0.000	0.000	1.000	39975	0.885
E_hier_lineage	0.009 [0.004, 0.015]	0.088 [0.050, 0.125]	0.088	0.046	0.158	0.954	0.033	0.559	0.680	0.847	69321	0.133
C_recursive_sum	0.008 [0.003, 0.014]	0.038 [0.022, 0.053]	0.038	0.000	0.040	1.000	0.000	0.000	0.000	0.954	42097	0.089
D_hier_nolineage	0.008 [0.004, 0.015]	0.075 [0.055, 0.095]	0.075	0.000	0.128	1.000	0.000	0.000	0.000	0.856	55834	0.133

10,000 users (329,730 records, 40.0 hidden patterns, 10 seeds)

architecture	AP	found	R@100	rare recall	evidence cov.	FDR	decoy acc.	indep. acc.	lineage	info loss	compute (cu)	privacy exp.
A2_chunked_ctx	0.069 [0.052, 0.086]	0.663 [0.630, 0.698]	0.182	0.581	1.000	0.955	0.362	0.607	0.657	0.004	2.07e+06	0.226

architecture	AP	found	R@100	rare recall	evidence cov.	FDR	decoy acc.	indep. acc.	lineage	info loss	compute (cu)	privacy exp.
A_flat_rag	0.062 [0.049, 0.075]	0.648 [0.605, 0.690]	0.160	0.487	0.982	0.956	0.365	0.728	0.670	0.049	1.09e+06	0.127
B2_map_reduce	0.051 [0.035, 0.070]	0.315 [0.273, 0.358]	0.165	0.122	0.422	0.970	0.227	0.771	0.540	0.644	1.99e+05	0.885
J_mycelic_verified	0.033 [0.020, 0.049]	0.385 [0.335, 0.430]	0.135	0.218	0.677	0.974	0.205	0.737	0.515	0.334	1.12e+06	0.147
B_long_context	0.030 [0.025, 0.036]	0.465 [0.410, 0.530]	0.158	0.152	0.728	0.948	0.155	0.660	0.718	0.384	1.09e+06	0.127
I_mycelic_completion	0.030 [0.019, 0.041]	0.380 [0.340, 0.422]	0.110	0.187	0.625	0.974	0.225	0.766	0.502	0.371	2.00e+06	0.147
H_mycelic_full	0.029 [0.016, 0.044]	0.385 [0.335, 0.430]	0.107	0.218	0.677	0.974	0.205	0.737	0.515	0.334	1.11e+06	0.147
G_hier_questions	0.029 [0.018, 0.040]	0.398 [0.342, 0.458]	0.120	0.228	0.695	0.973	0.215	0.769	0.522	0.317	1.11e+06	0.138
B4_central_triage	0.024 [0.016, 0.032]	0.388 [0.338, 0.440]	0.107	0.257	0.950	0.974	0.200	0.727	0.487	0.114	3.98e+05	0.885
Z_random_rank	0.013 [0.009, 0.016]	0.315 [0.273, 0.358]	0.065	0.122	0.422	0.970	0.227	0.771	0.540	0.644	1.99e+05	0.885
F_hier_retrieval	0.009 [0.005, 0.013]	0.100 [0.073, 0.130]	0.065	0.087	0.130	0.983	0.030	0.755	0.548	0.857	3.77e+05	0.138
Y_oracle_retrieval	0.009 [0.004, 0.015]	0.232 [0.188, 0.272]	0.048	0.148	1.000	0.984	0.133	0.527	0.460	0.004	4.87e+05	0.885
E_hier_lineage	0.004 [0.002, 0.008]	0.035 [0.022, 0.048]	0.035	0.000	0.065	0.982	0.005	0.443	0.567	0.924	2.27e+05	0.138
Z2_naive_enumerate	0.004 [0.001, 0.008]	0.143 [0.075, 0.227]	0.042	0.156	0.000	0.990	0.140	0.000	0.000	1.000	1.84e+05	0.885
D_hier_nolineage	0.002 [0.001, 0.004]	0.043 [0.033, 0.053]	0.043	0.000	0.057	1.000	0.000	0.000	0.000	0.935	1.96e+05	0.138
C_recursive_sum	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000	0.000	0.003	1.000	0.000	0.000	0.000	0.990	1.69e+05	0.092

50,000 users (1,639,365 records, 100.0 hidden patterns, 5 seeds)

architecture	AP	found	R@100	rare recall	evidence cov.	FDR	decoy acc.	indep. acc.	lineage	info loss	compute (cu)	privacy exp.
A2_chunked_ctx	0.051 [0.032, 0.082]	0.686 [0.658, 0.716]	0.112	0.608	1.000	0.977	0.368	0.594	0.641	0.003	1.02e+07	0.223
A_flat_rag	0.013 [0.010, 0.017]	0.424 [0.378, 0.474]	0.038	0.109	0.662	0.973	0.148	0.655	0.693	0.446	1.24e+06	0.025
I_mycelic_completion	0.012 [0.008, 0.015]	0.330 [0.292, 0.370]	0.040	0.182	0.404	0.988	0.202	0.751	0.491	0.586	5.09e+06	0.146
J_mycelic_verified	0.011 [0.008, 0.015]	0.336 [0.296, 0.376]	0.040	0.189	0.420	0.988	0.204	0.751	0.511	0.574	2.94e+06	0.146
H_mycelic_full	0.009 [0.008, 0.010]	0.336 [0.296, 0.376]	0.022	0.189	0.420	0.988	0.204	0.751	0.511	0.574	2.93e+06	0.146
B4_central_triage	0.008 [0.007, 0.011]	0.314 [0.302, 0.328]	0.024	0.112	0.462	0.981	0.282	0.683	0.484	0.602	1.07e+06	0.885
G_hier_questions	0.008 [0.008, 0.009]	0.336 [0.296, 0.376]	0.028	0.180	0.420	0.988	0.210	0.753	0.508	0.570	2.92e+06	0.138
B2_map_reduce	0.008 [0.001, 0.017]	0.056 [0.030, 0.082]	0.022	0.018	0.072	0.987	0.026	0.801	0.582	0.941	7.78e+05	0.885
Y_oracle_retrieval	0.005 [0.003, 0.007]	0.284 [0.220, 0.324]	0.020	0.220	1.000	0.991	0.148	0.541	0.477	0.003	2.43e+06	0.885
B_long_context	0.002 [0.001, 0.003]	0.076 [0.056, 0.098]	0.018	0.007	0.186	0.984	0.022	0.606	0.615	0.808	1.12e+06	0.025
Z2_naive_enumerate	0.001 [0.000, 0.002]	0.138 [0.086, 0.190]	0.006	0.125	0.000	0.995	0.150	0.000	0.000	1.000	9.21e+05	0.885
Z_random_rank	0.001 [0.000, 0.002]	0.056 [0.030, 0.082]	0.010	0.018	0.072	0.987	0.026	0.801	0.582	0.941	7.78e+05	0.885
F_hier_retrieval	0.001 [0.000, 0.001]	0.016 [0.008, 0.024]	0.012	0.005	0.028	0.993	0.004	0.625	0.426	0.961	1.15e+06	0.138
E_hier_lineage	0.001 [0.000, 0.001]	0.008 [0.004, 0.010]	0.008	0.000	0.018	0.989	0.000	0.230	0.400	0.976	9.90e+05	0.138
D_hier_nolineage	0.000 [0.000, 0.000]	0.002 [0.000, 0.006]	0.002	0.000	0.002	1.000	0.000	0.000	0.000	0.990	8.82e+05	0.138
C_recursive_sum	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000	0.000	0.002	1.000	0.000	0.000	0.000	0.996	8.05e+05	0.092

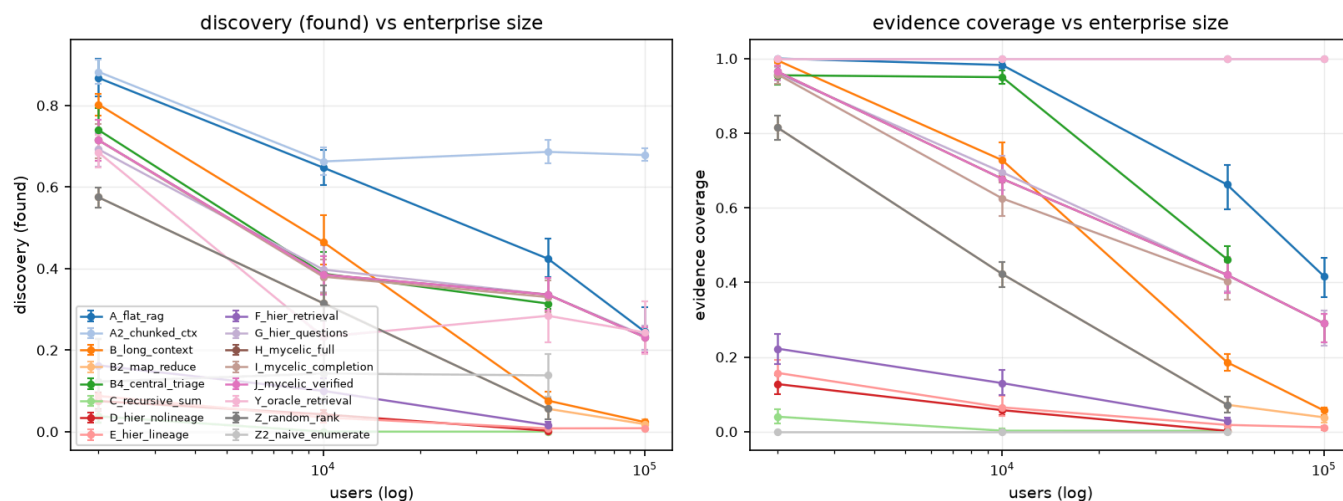
Paired comparisons

Every comparison is paired on (scale, seed) — the seed controls both the org chart and the corpus, so unpaired tests would be swamped by between-world variance.

comparison	metric	mean A	mean B	Δ	95% CI	wins	sign p
H_mycelic_full vs E_hier_lineage	found	0.5072	0.0506	+0.4566	[+0.3938, +0.5240]	25/25	0.000
H_mycelic_full vs B2_map_reduce	found	0.5072	0.3672	+0.1400	[+0.0978, +0.1824]	22/23	0.000
H_mycelic_full vs A_flat_rag	found	0.5072	0.6908	-0.1836	[-0.2276, -0.1388]	1/25	0.000
H_mycelic_full vs B_long_context	found	0.5072	0.5222	-0.0150	[-0.0724, +0.0486]	8/24	0.152
H_mycelic_full vs C_recursive_sum	found	0.5072	0.0150	+0.4922	[+0.4270, +0.5602]	25/25	0.000
H_mycelic_full vs D_hier_nolineage	found	0.5072	0.0474	+0.4598	[+0.3960, +0.5276]	25/25	0.000
H_mycelic_full vs B4_central_triage	found	0.5072	0.5138	-0.0066	[-0.0410, +0.0254]	12/25	1.000
H_mycelic_full vs Y_oracle_retrieval	found	0.5072	0.4238	+0.0834	[+0.0394, +0.1258]	19/23	0.003

comparison	metric	mean A	mean B	Δ	95% CI	wins	sign p
B2_map_reduce vs Z_random_rank	found	0.3672	0.3672	+0.0000	[+0.0000, +0.0000]	0/0	1.000
G_hier_questions vs F_hier_retrieval	found	0.5032	0.1082	+0.3950	[+0.3398, +0.4522]	25/25	0.000
F_hier_retrieval vs E_hier_lineage	found	0.1082	0.0506	+0.0576	[+0.0378, +0.0784]	21/22	0.000
H_mycelic_full vs E_hier_lineage	AP	0.0431	0.0056	+0.0375	[+0.0256, +0.0509]	23/25	0.000
H_mycelic_full vs B2_map_reduce	AP	0.0431	0.0537	-0.0106	[-0.0204, -0.0013]	10/25	0.424
H_mycelic_full vs A_flat_rag	AP	0.0431	0.1256	-0.0825	[-0.1205, -0.0495]	2/25	0.000
H_mycelic_full vs B_long_context	AP	0.0431	0.0837	-0.0406	[-0.0684, -0.0164]	10/25	0.424
H_mycelic_full vs C_recursive_sum	AP	0.0431	0.0033	+0.0398	[+0.0281, +0.0529]	25/25	0.000
H_mycelic_full vs D_hier_nolineage	AP	0.0431	0.0043	+0.0388	[+0.0271, +0.0525]	25/25	0.000
H_mycelic_full vs B4_central_triage	AP	0.0431	0.0480	-0.0049	[-0.0155, +0.0050]	12/25	1.000
H_mycelic_full vs Y_oracle_retrieval	AP	0.0431	0.0262	+0.0169	[+0.0099, +0.0248]	22/25	0.000
B2_map_reduce vs Z_random_rank	AP	0.0537	0.0219	+0.0318	[+0.0217, +0.0422]	25/25	0.000
G_hier_questions vs F_hier_retrieval	AP	0.0385	0.0131	+0.0254	[+0.0176, +0.0340]	23/25	0.000
F_hier_retrieval vs E_hier_lineage	AP	0.0131	0.0056	+0.0075	[+0.0030, +0.0125]	19/24	0.007

11. Scale



architecture	2,000	10,000	50,000	100,000
A_flat_rag	0.867	0.648	0.424	0.245
A2_chunked_ctx	0.882	0.663	0.686	0.678
B_long_context	0.802	0.465	0.076	0.023
B2_map_reduce	0.575	0.315	0.056	0.018
B4_central_triage	0.740	0.388	0.314	—
E_hier_lineage	0.088	0.035	0.008	0.008
G_hier_questions	0.693	0.398	0.336	0.235
H_mycelic_full	0.715	0.385	0.336	0.232
J_mycelic_verified	0.715	0.385	0.336	0.232
Y_oracle_retrieval	0.685	0.232	0.284	0.243

Compute (cu) at the same points:

architecture	2,000	10,000	50,000	100,000
A_flat_rag	4.71e+05	1.09e+06	1.24e+06	1.41e+06

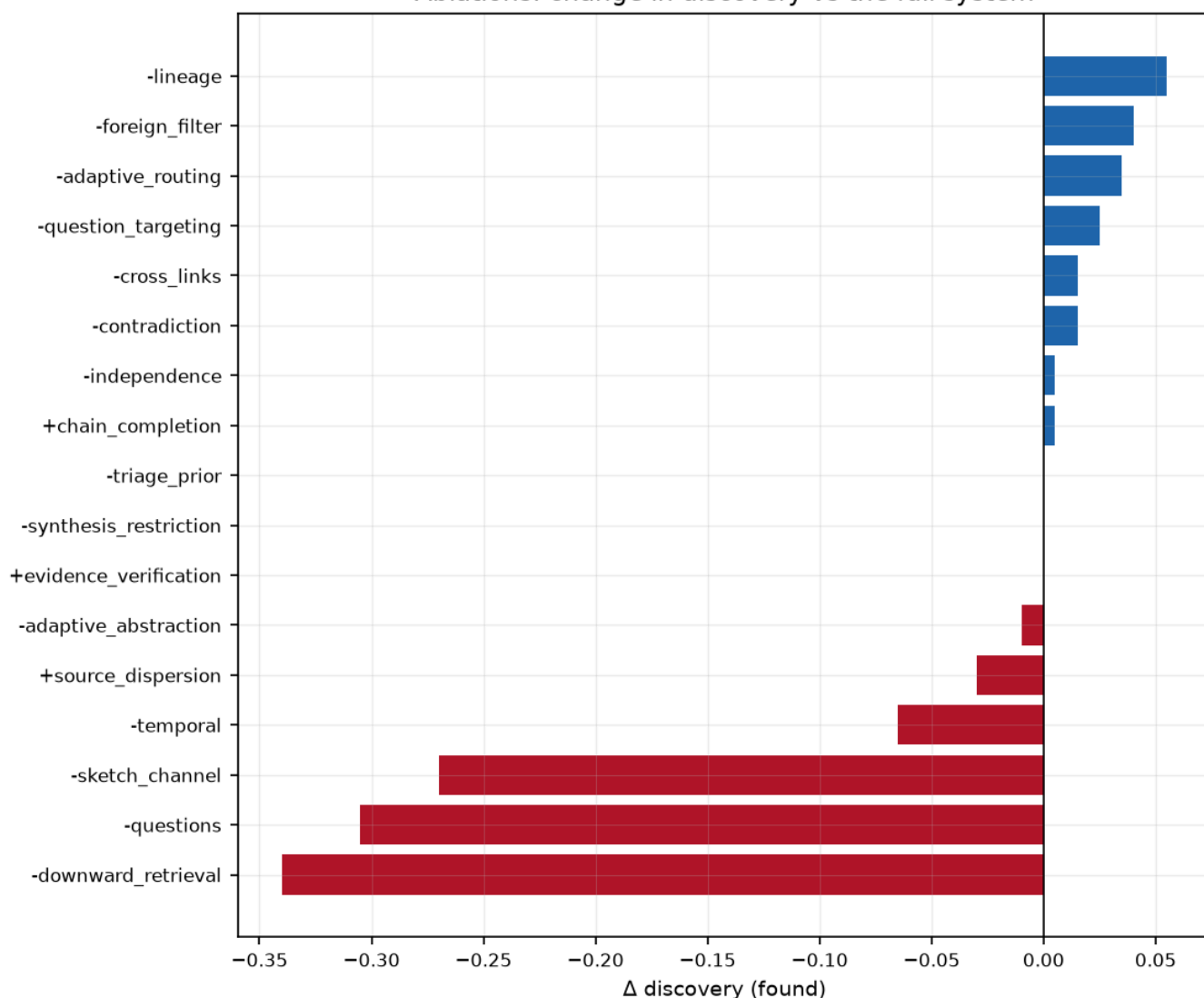
architecture	2,000	10,000	50,000	100,000
A2_chunked_ctx	5.12e+05	2.07e+06	1.02e+07	2.03e+07
B_long_context	1.09e+06	1.09e+06	1.12e+06	1.11e+06
B2_map_reduce	7.97e+04	1.99e+05	7.78e+05	1.49e+06
B4_central_triage	1.33e+05	3.98e+05	1.07e+06	—
E_hier_lineage	6.93e+04	2.27e+05	9.90e+05	1.92e+06
G_hier_questions	8.37e+05	1.11e+06	2.92e+06	4.51e+06
H_mycelic_full	8.43e+05	1.11e+06	2.93e+06	4.71e+06
J_mycelic_verified	8.53e+05	1.12e+06	2.94e+06	4.72e+06
Y_oracle_retrieval	1.44e+05	4.87e+05	2.43e+06	4.86e+06

Does the picture change with size? From 2,000 to 100,000 users — 50x the people — the hierarchy's discovery goes $0.715 \rightarrow 0.232$ and A2_chunked_ctx 's goes $0.882 \rightarrow 0.678$, so the gap widens rather than closing. Compute tells the opposite story: the hierarchy's rises 5.6x over that 50x growth in people, A2_chunked_ctx 's 39.8x, so the hierarchy's cost advantage grows with size:

A2_chunked_ctx costs 0.6x the hierarchy at 2,000 users — i.e. it is the cheaper of the two there — and 4.3x at 100,000. Neither curve crosses inside the range measured. The trend is favourable to the hierarchy on cost and unfavourable on accuracy, and nothing here justifies extrapolating a crossing point beyond 100,000 — that would be an extrapolation, and this study does not make any.

12. Ablations

Ablations: change in discovery vs the full system



variant	AP	found	rare recall	evidence cov.	FDR	indep. acc.	lineage	contra F1	D4 support infl.	Q utility	compute (cu)
-lineage	0.035	0.430 [0.405, 0.455]	0.084	0.665	0.952	0.754	0.000	0.210	2.912	0.835	1.05e+06
-foreign_filter	0.028	0.415 [0.370, 0.470]	0.273	0.690	0.972	0.742	0.463	0.111	3.124	0.906	1.23e+06
+chain_completion	0.028	0.380 [0.330, 0.445]	0.210	0.630	0.974	0.745	0.474	0.263	3.319	0.897	1.95e+06
-cross_links	0.027	0.390 [0.345, 0.445]	0.227	0.700	0.974	0.763	0.507	0.000	3.017	0.908	1.08e+06
+evidence_verification	0.027	0.375 [0.315, 0.435]	0.215	0.690	0.975	0.733	0.492	0.191	2.832	0.897	1.10e+06
-contradiction	0.027	0.390 [0.345, 0.430]	0.255	0.690	0.974	0.747	0.504	0.267	3.106	0.906	1.08e+06
-question_targeting	0.026	0.400 [0.355, 0.445]	0.253	0.685	0.973	0.760	0.504	0.120	2.998	0.891	1.09e+06
-adaptive_routing	0.026	0.410 [0.355, 0.465]	0.273	0.685	0.972	0.759	0.497	0.380	2.937	0.905	1.11e+06
-triage_prior	0.023	0.375 [0.315, 0.435]	0.215	0.690	0.975	0.733	0.492	0.191	2.832	0.897	1.09e+06
full	0.023	0.375 [0.315, 0.435]	0.215	0.690	0.975	0.733	0.492	0.191	2.832	0.897	1.09e+06
-synthesis_restriction	0.021	0.375 [0.330, 0.425]	0.211	0.695	0.975	0.745	0.486	0.280	2.919	0.896	1.09e+06
+source_dispersion	0.020	0.345 [0.285, 0.405]	0.263	0.695	0.977	0.738	0.523	0.102	2.701	0.916	1.09e+06
-adaptive_abstraction	0.018	0.365 [0.310, 0.415]	0.210	0.665	0.975	0.762	0.505	0.191	2.983	0.905	1.08e+06
-independence	0.018	0.380 [0.305, 0.450]	0.278	0.685	0.974	0.741	0.505	0.275	3.974	0.909	1.08e+06
-temporal	0.010	0.310 [0.260, 0.365]	0.205	0.680	0.979	0.766	0.478	0.150	3.433	0.888	1.10e+06
-sketch_channel	0.005	0.105 [0.080, 0.130]	0.150	0.120	0.989	0.693	0.514	0.000	2.757	0.998	1.22e+06
-questions	0.003	0.070 [0.045, 0.095]	0.048	0.075	0.986	0.684	0.498	0.000	2.000	0.000	3.75e+05
-downward_retrieval	0.002	0.035 [0.015, 0.055]	0.013	0.060	0.982	0.518	0.598	0.000	1.133	0.000	2.30e+05

Paired against the full system (same seeds):

removed	Δ found	95% CI	wins	sign p	Δ AP	Δ compute
+chain_completion	+0.0050	[-0.0300, +0.0350]	2/5	1.000	+0.0055	+8.570e+05
+evidence_verification	-0.0000	[-0.0000, -0.0000]	0/0	1.000	+0.0041	+1.031e+04
+source_dispersion	-0.0300	[-0.0500, -0.0100]	3/3	0.250	-0.0029	-1.689e+03
-adaptive_abstraction	-0.0100	[-0.0700, +0.0450]	2/5	1.000	-0.0043	-5.201e+03
-adaptive_routing	+0.0350	[-0.0101, +0.0850]	1/4	0.625	+0.0029	+1.827e+04
-contradiction	+0.0150	[-0.0400, +0.0750]	1/3	1.000	+0.0040	-5.620e+03
-cross_links	+0.0150	[-0.0300, +0.0600]	2/5	1.000	+0.0044	-9.696e+03
-downward_retrieval	-0.3400	[-0.3950, -0.2900]	5/5	0.062	-0.0203	-8.584e+05
-foreign_filter	+0.0400	[-0.0250, +0.0950]	1/5	0.375	+0.0058	+1.410e+05
-independence	+0.0050	[-0.0600, +0.0550]	1/4	0.625	-0.0047	-8.090e+03
-lineage	+0.0550	[+0.0000, +0.1100]	1/4	0.625	+0.0125	-4.012e+04
-question_targeting	+0.0250	[-0.0200, +0.0800]	1/4	0.625	+0.0037	+3.213e+02
-questions	-0.3050	[-0.3550, -0.2650]	5/5	0.062	-0.0193	-7.129e+05
-sketch_channel	-0.2700	[-0.3400, -0.2150]	5/5	0.062	-0.0176	+1.311e+05
-synthesis_restriction	-0.0000	[-0.0200, +0.0250]	2/3	1.000	-0.0013	+1.270e+03
-temporal	-0.0650	[-0.1100, -0.0250]	4/4	0.125	-0.0129	+7.386e+03
-triage_prior	-0.0000	[-0.0000, -0.0000]	0/0	1.000	-0.0000	-0.000e+00

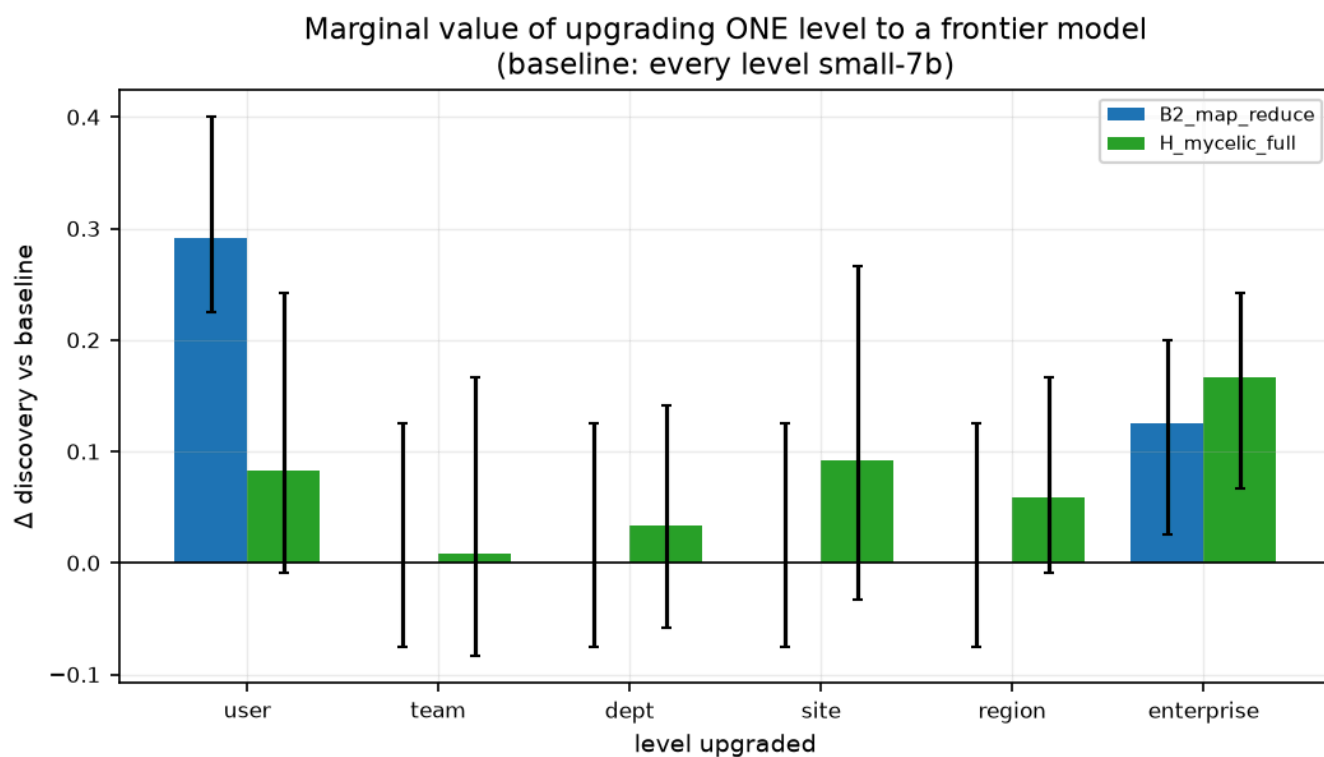
Reading the ablations. Δ is the variant minus the full system. For a - row, a negative Δ means removing the mechanism made things worse, so the mechanism is doing work. For a + row the mechanism is *added* to the full system, so a positive Δ is the case for adopting it. A CI spanning zero means the seed count cannot separate the two; with 5 seeds an exact sign test cannot go below 0.0625, so the interval is carrying more of the argument than the p-value.

- Load-bearing** — removing it costs discovery, interval entirely below zero: -downward_retrieval (-0.340), -questions (-0.305), -sketch_channel (-0.270), -temporal (-0.065).
- Not distinguishable from noise at this seed count:** -adaptive_abstraction, -adaptive_routing, -contradiction, -cross_links, -foreign_filter, -independence, -lineage, -question_targeting, -synthesis_restriction, -

`trriage_prior`. These are not shown to be useless; they are shown to be unmeasured, which is a different statement and the honest one.

- **Variants tested on top of the full system.** These are not "the feature off vs on" — the full system already carries whatever calibration selected — so each is shown with the setting it actually changes:
 - `+chain_completion = chain_completion=True` → $\Delta +0.005$ $[-0.030, +0.035]$ — inconclusive at this seed count
 - `+evidence_verification = verify_evidence=True` → $\Delta -0.000$ $[-0.000, -0.000]$ — inconclusive at this seed count
 - `+source_dispersion = w_dispersion=1.5` → $\Delta -0.030$ $[-0.050, -0.010]$ — **do not adopt**

13. Where compute should go



Marginal value of capability at each level

Baseline: every level running the same small model. Then exactly one level is upgraded to a frontier model, holding everything else fixed. This is the experiment that decides where model budget should go.

B2_map_reduce — baseline is every level at small-7b

level upgraded to frontier	AP	found	Δ found	compute	Δ compute	Δfound / Mcu
user	0.0451	0.442	0.292	4.35e+06	4.20e+06	0.0694
enterprise	0.0422	0.275	0.125	1.72e+05	26494	0.1250
dept	0.0058	0.150	0.000	1.46e+05	0	0.0000
none	0.0058	0.150	0.000	1.46e+05	0	0.0000
region	0.0058	0.150	0.000	1.46e+05	0	0.0000
site	0.0058	0.150	0.000	1.46e+05	0	0.0000
team	0.0058	0.150	0.000	1.46e+05	0	0.0000

level	Δ found	95% CI	wins	sign p
dept	+0.0000	[+0.0000, +0.0000]	0/0	1.000
enterprise	+0.1250	[+0.0750, +0.2000]	3/3	0.250
region	+0.0000	[+0.0000, +0.0000]	0/0	1.000

level	Δ found	95% CI	wins	sign p
site	+0.0000	[+0.0000, +0.0000]	0/0	1.000
team	+0.0000	[+0.0000, +0.0000]	0/0	1.000
user	+0.2917	[+0.2750, +0.3000]	3/3	0.250

Where the model budget goes for B2_map_reduce . Upgrading the **user** level buys the most discovery (+0.292) and upgrading **enterprise** buys the most per unit of extra compute (0.125 discovery per Mcu). The **team** level buys the least (+0.000). Ranked: user +0.292 > enterprise +0.125 > team +0.000 > dept +0.000 > site +0.000 > region +0.000. The pattern is **only the top level matters at all** — every intermediate level is flat at zero — and note that the user level is doing extraction from raw text, a different job from the aggregation the middle levels do, so its gain does not belong to the same trend as theirs. Every interval here is from 3 runs per cell, so the ordering is directional and the individual gaps are not separable.

H_mycelic_full — baseline is every level at small-7b

level upgraded to frontier	AP	found	Δ found	compute	Δ compute	Δ found / Mcu
enterprise	0.0175	0.400	0.167	5.75e+05	2.54e+05	0.1667
site	0.0152	0.325	0.092	5.79e+05	2.58e+05	0.0917
user	0.0150	0.317	0.083	5.62e+06	5.29e+06	0.0157
region	0.0111	0.292	0.058	4.21e+05	1.01e+05	0.0583
dept	0.0106	0.267	0.033	1.22e+06	9.04e+05	0.0333
team	0.0062	0.242	0.008	2.54e+06	2.22e+06	0.0038
none	0.0077	0.233	0.000	3.20e+05	0	0.0000

level	Δ found	95% CI	wins	sign p
dept	+0.0333	[+0.0000, +0.0750]	2/2	0.500
enterprise	+0.1667	[+0.0750, +0.2250]	3/3	0.250
region	+0.0583	[+0.0000, +0.1250]	2/2	0.500
site	+0.0917	[+0.0250, +0.1500]	3/3	0.250
team	+0.0083	[-0.1000, +0.0750]	2/3	1.000
user	+0.0833	[-0.0250, +0.1500]	2/3	1.000

Where the model budget goes for H_mycelic_full . Upgrading the **enterprise** level buys the most discovery (+0.167) and upgrading **enterprise** buys the most per unit of extra compute (0.167 discovery per Mcu). The **team** level buys the least (+0.008). Ranked: enterprise +0.167 > site +0.092 > user +0.083 > region +0.058 > dept +0.033 > team +0.008. The pattern is **not a clean trend** — and note that the user level is doing extraction from raw text, a different job from the aggregation the middle levels do, so its gain does not belong to the same trend as theirs. Every interval here is from 3 runs per cell, so the ordering is directional and the individual gaps are not separable.

Allocation across levels

B2_map_reduce

Best discovery: **flat-frontier** (0.592 at 4.37e+06 cu). Best value: **lexical-bottom** (0.300 at 5.39e+04 cu — 5.02e+03 cu per correct discovery, against 1.91e+05 for the best-discovery option). **lexical-bottom** keeps 51% of the discovery for 1% of the compute. Whether that trade is worth making is a budget question, not a research one — but it is the answer to "should capability increase up the ladder?": a *graduated* ladder is not what wins here. What wins is a cheap edge and a strong kernel, because the kernel is where the discrimination happens and the edge is only extracting. For reference the cheapest allocation of all, **lexical-bottom** , reaches 0.300 at 5.39e+04 cu, so the spread between doing nothing clever and doing the most expensive thing is +0.292 discovery for 81x the compute.

allocation	user → ... → kernel	AP	found	rare	compute	cu/disc	wall s
flat-frontier	frontier->frontier->frontier->frontier->frontier->frontier	0.1446	0.592	0.204	4.37e+06	1.91e+05	238.7
flat-mid	mid-32b->mid-32b->mid-32b->mid-32b->mid-32b->mid-32b	0.0556	0.392	0.231	6.83e+05	45684	94.6

allocation	user → ... → kernel	AP	found	rare	compute	cu/disc	wall s
front-loaded	mid-14b->mid-32b->mid-32b->mid-14b->mid-14b->large-70b	0.0522	0.342	0.204	3.03e+05	23280	115.7
lexical-bottom	lexical->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0736	0.300	0.184	53852	5016	223.8
lexical-to-kernel	lexical->lexical->lexical->lexical->lexical->frontier-plus	0.0736	0.300	0.184	53852	5016	223.8
back-loaded	small-7b->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0401	0.292	0.081	1.99e+05	18498	239.4
naive-ladder	small-7b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0401	0.292	0.081	1.99e+05	18498	239.4
two-step	small-7b->small-7b->mid-32b->mid-32b->mid-32b->frontier-plus	0.0401	0.292	0.081	1.99e+05	18498	239.4
edge-bottom	edge-3b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0101	0.275	0.203	1.21e+05	11760	272.5
kernel-only	edge-3b->edge-3b->edge-3b->edge-3b->edge-3b->frontier-plus	0.0101	0.275	0.203	1.21e+05	11760	272.5
flat-small	small-7b->small-7b->small-7b->small-7b->small-7b->small-7b	0.0058	0.150	0.061	1.46e+05	41853	60.7

E_hier_lineage

Best discovery: **flat-frontier** (0.300 at 5.38e+06 cu). Best value: **lexical-to-kernel** (0.067 at 1.47e+04 cu — 9.92e+03 cu per correct discovery, against 6.76e+05 for the best-discovery option). **lexical-to-kernel** keeps 22% of the discovery for 0% of the compute. Whether that trade is worth making is a budget question, not a research one — but it is the answer to "should capability increase up the ladder?": a *graduated* ladder is not what wins here. What wins is a cheap edge and a strong kernel, because the kernel is where the discrimination happens and the edge is only extracting. For reference the cheapest allocation of all, **lexical-to-kernel**, reaches 0.067 at 1.47e+04 cu, so the spread between doing nothing clever and doing the most expensive thing is +0.233 discovery for 365x the compute.

allocation	user → ... → kernel	AP	found	rare	compute	cu/disc	wall s
flat-frontier	frontier->frontier->frontier->frontier->frontier->frontier	0.0897	0.300	0.081	5.38e+06	6.76e+05	260.2
flat-mid	mid-32b->mid-32b->mid-32b->mid-32b->mid-32b->mid-32b	0.0078	0.100	0.022	8.31e+05	4.38e+05	92.0
lexical-to-kernel	lexical->lexical->lexical->lexical->lexical->frontier-plus	0.0206	0.067	0.000	14746	9922	47.1
front-loaded	mid-14b->mid-32b->mid-32b->mid-14b->mid-14b->large-70b	0.0045	0.050	0.000	4.37e+05	3.64e+05	81.4
lexical-bottom	lexical->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0043	0.050	0.000	82671	41335	124.2
two-step	small-7b->small-7b->mid-32b->mid-32b->mid-32b->frontier-plus	0.0043	0.042	0.000	2.22e+05	1.85e+05	105.0
edge-bottom	edge-3b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0006	0.033	0.022	1.95e+05	1.95e+05	152.2
naive-ladder	small-7b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0046	0.033	0.022	2.79e+05	2.16e+05	144.2
back-loaded	small-7b->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0059	0.025	0.000	2.24e+05	1.86e+05	129.8
flat-small	small-7b->small-7b->small-7b->small-7b->small-7b->small-7b	0.0005	0.017	0.022	1.73e+05	1.73e+05	52.9
kernel-only	edge-3b->edge-3b->edge-3b->edge-3b->edge-3b->frontier-plus	0.0000	0.000	0.000	80415	80415	53.3

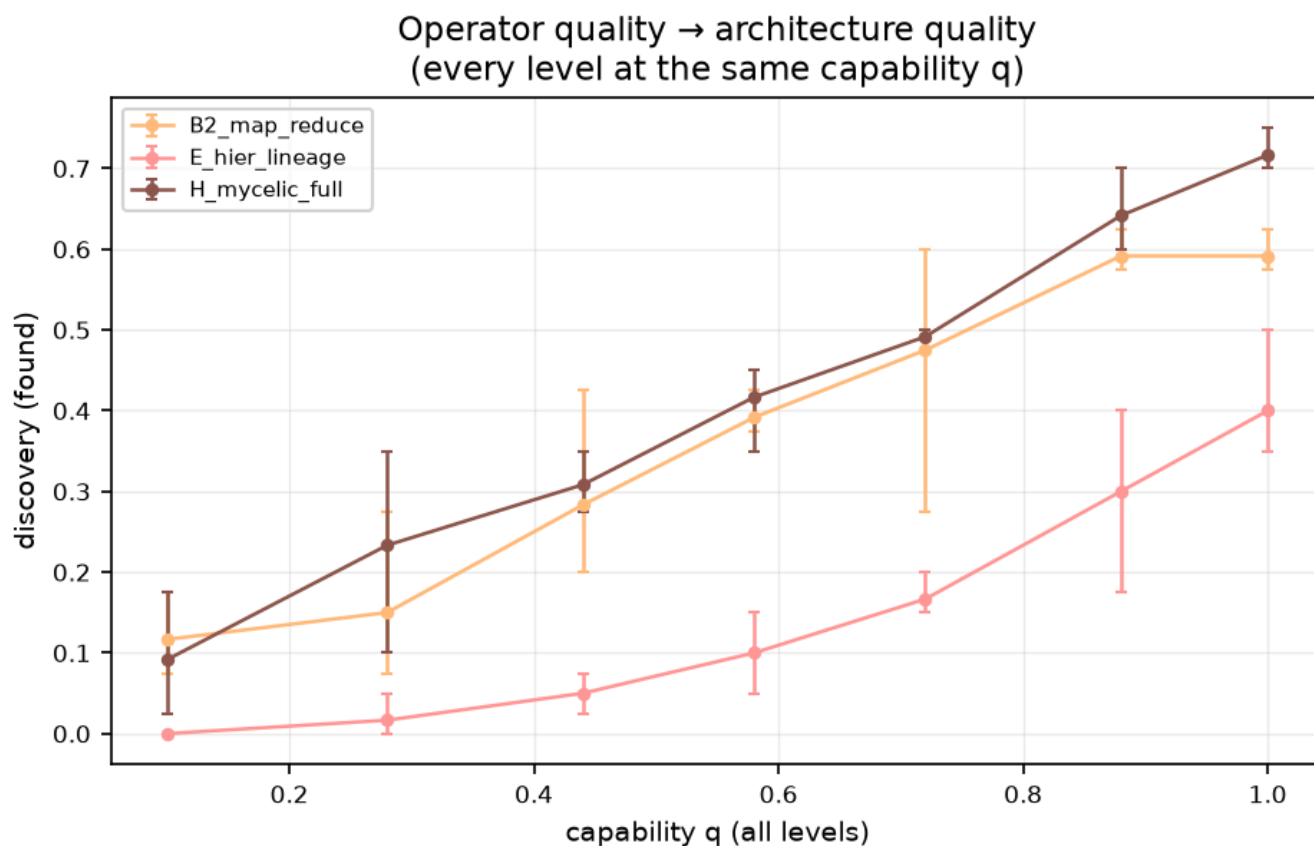
H_mycelic_full

Best discovery: **flat-frontier** (0.642 at 9.55e+06 cu). Best value: **lexical-to-kernel** (0.408 at 5.08e+05 cu — 3.24e+04 cu per correct discovery, against 3.83e+05 for the best-discovery option). **lexical-to-kernel** keeps 64% of the discovery for 5% of the compute. Whether that trade is worth making is a budget question, not a research one — but it is the answer to "should capability increase up the ladder?": a *graduated* ladder is not what wins here. What wins is a cheap edge and a strong kernel, because the kernel is where the discrimination happens and the edge is only extracting. For reference the cheapest allocation of all, **flat-small**, reaches 0.233 at 3.20e+05 cu, so the spread between doing nothing clever and doing the most expensive thing is +0.408 discovery for 30x the compute.

allocation	user → ... → kernel	AP	found	rare	compute	cu/disc	wall s
flat-frontier	frontier->frontier->frontier->frontier->frontier->frontier	0.0668	0.642	0.454	9.55e+06	3.83e+05	1025.9
flat-mid	mid-32b->mid-32b->mid-32b->mid-32b->mid-32b->mid-32b	0.0262	0.417	0.284	1.46e+06	88757	441.9
lexical-to-kernel	lexical->lexical->lexical->lexical->lexical->frontier-plus	0.0346	0.408	0.250	5.08e+05	32356	1000.2
naive-ladder	small-7b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0199	0.400	0.217	1.33e+06	83309	1409.7
front-loaded	mid-14b->mid-32b->mid-32b->mid-14b->mid-14b->large-70b	0.0142	0.392	0.254	1.00e+06	67449	522.3
two-step	small-7b->small-7b->mid-32b->mid-32b->mid-32b->frontier-plus	0.0171	0.375	0.120	1.07e+06	72053	1304.9

allocation	user → ... → kernel	AP	found	rare	compute	cu/disc	wall s
back-loaded	small-7b->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0248	0.342	0.178	1.08e+06	81105	1347.3
lexical-bottom	lexical->small-7b->mid-14b->mid-32b->frontier->frontier-plus	0.0213	0.342	0.267	9.04e+05	69761	1360.1
edge-bottom	edge-3b->mid-14b->mid-32b->large-70b->frontier->frontier-plus	0.0070	0.267	0.156	1.25e+06	1.18e+05	1433.4
kernel-only	edge-3b->edge-3b->edge-3b->edge-3b->edge-3b->frontier-plus	0.0092	0.250	0.167	6.58e+05	69921	1128.2
flat-small	small-7b->small-7b->small-7b->small-7b->small-7b->small-7b	0.0077	0.233	0.198	3.20e+05	45522	280.4

Uniform capability sweep



The function from operator quality to architecture quality — which is the claim this study can actually make, since named model classes are assumed positions on this axis rather than measured ones.

architecture	q	AP	found	rare	FDR	compute
B2_map_reduce	0.10	0.0015	0.117	0.053	0.992	89103
B2_map_reduce	0.28	0.0058	0.150	0.061	0.985	1.75e+05
B2_map_reduce	0.44	0.0338	0.283	0.117	0.969	3.21e+05
B2_map_reduce	0.58	0.0556	0.392	0.231	0.955	5.44e+05
B2_map_reduce	0.72	0.0444	0.475	0.237	0.946	9.22e+05
B2_map_reduce	0.88	0.1446	0.592	0.204	0.935	1.68e+06
B2_map_reduce	1.00	0.1015	0.592	0.223	0.943	2.64e+06
E_hier_lineage	0.10	0.0000	0.000	0.000	1.000	1.04e+05
E_hier_lineage	0.28	0.0005	0.017	0.022	0.994	2.09e+05
E_hier_lineage	0.44	0.0028	0.050	0.020	0.979	3.87e+05
E_hier_lineage	0.58	0.0078	0.100	0.022	0.949	6.61e+05
E_hier_lineage	0.72	0.0195	0.167	0.000	0.904	1.13e+06
E_hier_lineage	0.88	0.0897	0.300	0.081	0.749	2.07e+06

architecture	q	AP	found	rare	FDR	compute
E_hier_lineage	1.00	0.2341	0.400	0.159	0.483	3.25e+06
H_mycelic_full	0.10	0.0012	0.092	0.081	0.994	1.98e+05
H_mycelic_full	0.28	0.0077	0.233	0.198	0.984	3.86e+05
H_mycelic_full	0.44	0.0138	0.308	0.201	0.979	6.99e+05
H_mycelic_full	0.58	0.0262	0.417	0.284	0.972	1.16e+06
H_mycelic_full	0.72	0.0304	0.492	0.351	0.967	1.98e+06
H_mycelic_full	0.88	0.0668	0.642	0.454	0.950	3.68e+06
H_mycelic_full	1.00	0.0968	0.717	0.588	0.939	5.80e+06

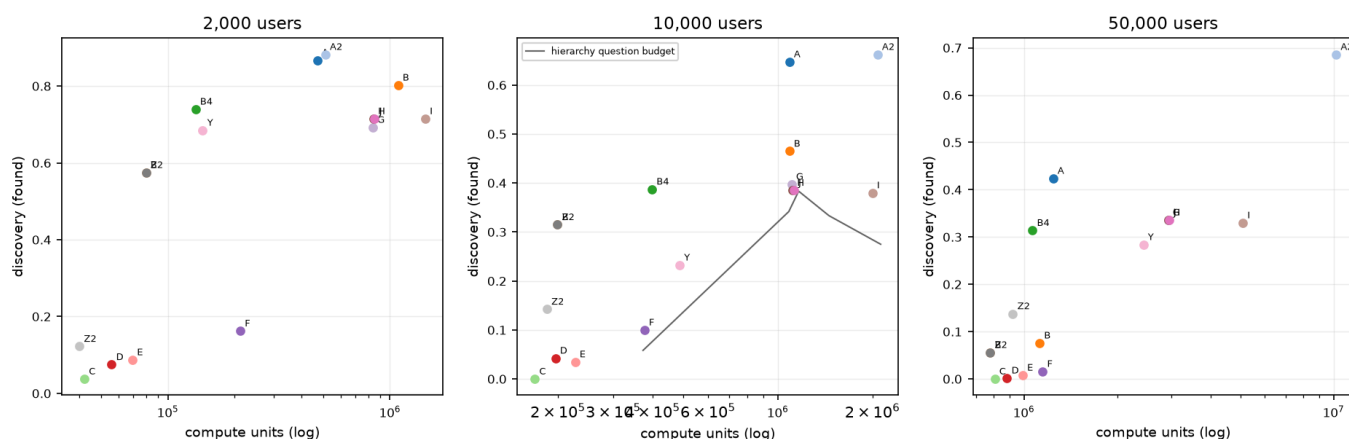
What the sweep says. Each architecture is run with *every* level set to the same capability `q`, so the curve isolates the operator from the topology.

architecture	found at q=0.10	found at q=1.00	gain	compute multiple	gain per doubling of compute
B2_map_reduce	0.117	0.592	+0.475	29.6x	+0.097
E_hier_lineage	0.000	0.400	+0.400	31.2x	+0.081
H_mycelic_full	0.092	0.717	+0.625	29.3x	+0.128

Capability is not worth the same everywhere. Per doubling of compute spent on better operators, `H_mycelic_full` converts it into +0.128 discovery and `E_hier_lineage` into +0.081. Read together with §13, which upgrades one level at a time: this table says how much a *uniform* capability increase is worth, §13 says where to put it if you are only buying one.

The row worth dwelling on is `E_hier_lineage`, pure upward aggregation with no descent. It finds 0.0% at $q=0.10$ and 40.0% at $q=1.00$. Its failure at realistic operator quality is therefore not a structural impossibility — a *perfect* summariser would make it work — but every real operator sits far enough below perfect that the structure cannot be rescued by a better model. That is a stronger negative result than 'it does not work', and a more useful one: it says the design is sensitive to exactly the thing we cannot guarantee.

14. Cost and quality



Hierarchy: question / descent budget

question budget frac	AP	found	rare	compute	cu/disc
0.00	0.0018	0.058	0.036	3.73e+05	2.27e+05
0.05	0.0248	0.342	0.178	1.08e+06	81105
0.15	0.0248	0.342	0.178	1.08e+06	81105
0.35	0.0233	0.383	0.287	1.16e+06	76226
0.55	0.0153	0.333	0.251	1.45e+06	1.10e+05
1.00	0.0113	0.275	0.198	2.12e+06	2.05e+05

The curve turns over. Discovery peaks at **0.35 question-budget fraction** (0.383) and falls to 0.275 at 1.0 — a loss of 0.108 from spending *more*. Cost per correct discovery is lowest at 0.35 (7.62e+04 cu). This is the study's central structural claim in one table: past a point, additional undifferentiated evidence crowds the genuine candidates down the ranking faster than it adds new ones. A budget is a filter, and removing the filter is not free.

Map-reduce: kernel object budget

kernel objects	AP	found	compute	cu/disc
300	0.0054	0.033	1.51e+05	1.18e+05
900	0.0310	0.108	1.64e+05	41083
2500	0.0401	0.292	1.99e+05	18498
8000	0.0067	0.267	2.66e+05	25566
20000	0.0031	0.225	3.91e+05	44955

The curve turns over. Discovery peaks at **2500 kernel objects** (0.292) and falls to 0.225 at 20000 — a loss of 0.067 from spending *more*. Cost per correct discovery is lowest at 2500 (1.85e+04 cu). This is the study's central structural claim in one table: past a point, additional undifferentiated evidence crowds the genuine candidates down the ranking faster than it adds new ones. A budget is a filter, and removing the filter is not free.

15. Latency

Latency is a critical-path estimate over the token counts actually metered: stages run in sequence, calls within a stage run in parallel up to a per-tier fleet concurrency. Per-architecture wall-clock and per-call p50/p95 appear in the §10 tables. The operationally important difference is not the estimate but the **call count**: the centralised designs make one to a few dozen model calls; the hierarchy makes tens of thousands. That is a real deployment difference in scheduling, retry and failure-domain terms, independent of how accurate the latency model is.

16. Information loss

information loss is the fraction of the hidden patterns' claim types that do not survive to the kernel's working pool. **evidence coverage** is the fraction of patterns for which the kernel ever held two or more of the pattern's links. The gap between evidence coverage and discovery is loss to **judgement**; the gap between 1.0 and evidence coverage is loss to **retrieval**. Both are in §10, and the distinction is the single most useful diagnostic in this study.

17. Continual questioning

The full system asks 220 questions per run. 90% of them change a conclusion — a question counts as useful only if the kernel's hypothesis set or its confidence actually moved, never merely because text was produced. Each returns 102 new knowledge objects on average, and 100% are well-targeted.

That last figure is not a measurement and should not be read as one. **question_targeting** is the fraction of questions the kernel aimed at the entity it was actually reasoning about, and in the simulator that is drawn directly from the kernel tier's **question_quality** parameter — so with a frontier kernel it is ~100% *by construction*, and it would report ~100% even if targeting were worthless. The informative test is the ablation below that destroys targeting outright, and that one does not reach significance. It is listed here because leaving a parameter echoed back as a headline number would be the kind of thing this report is supposed to catch.

- **removing questioning entirely**: discovery Δ -0.305 (95% CI [-0.355, -0.265], worse on 5/5 seeds, sign $p=0.062$), rare-signal recall Δ -0.166 (95% CI [-0.242, -0.090], worse on 5/5 seeds, sign $p=0.062$), compute -7.13e+05 cu.
- **keeping questions but destroying their targeting**: discovery Δ +0.025 (95% CI [-0.020, +0.080], worse on 1/4 seeds, sign $p=0.625$), rare-signal recall Δ +0.039 (95% CI [-0.020, +0.111], worse on 1/3 seeds, sign $p=1.000$), compute +3.21e+02 cu.

Every Δ above is *the ablated system minus the full one*, so a negative number means removing the mechanism made things worse. The seed count is the number of paired seeds on which the ablation was the worse of the two.

18. Downward retrieval, and adversarial conditions

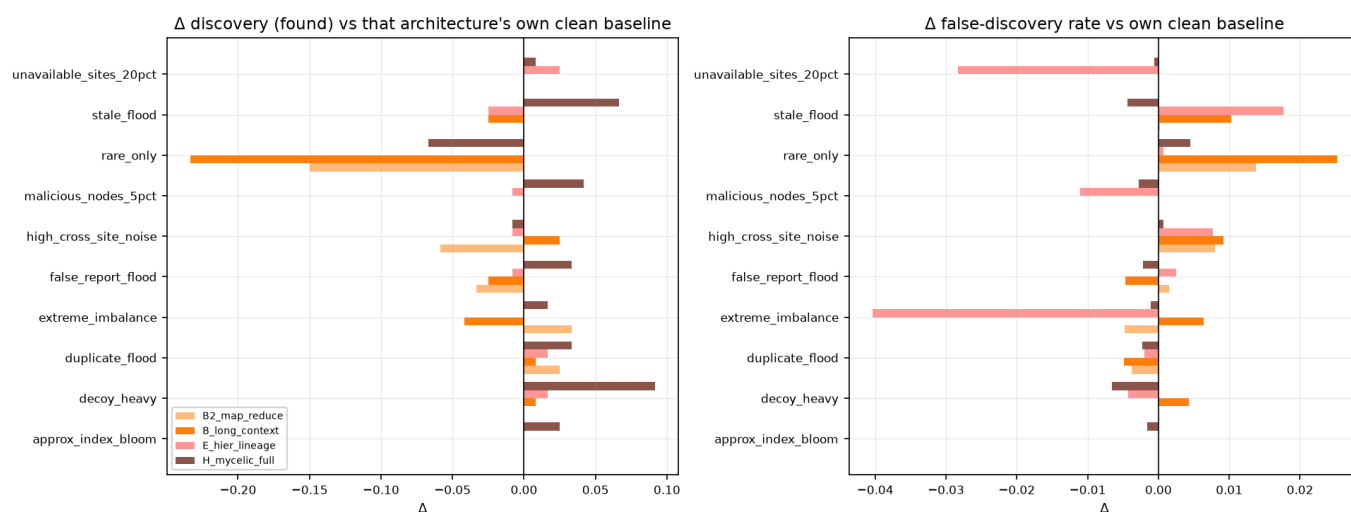
Downward retrieval

The downward path is the largest single contributor to the hierarchy's performance, and it works in two separable stages:

- **targeted descent** (E to F) — the kernel forms a hypothesis and routes a query down the index to the branches holding the missing evidence. Expansion is best-first with a per-parent quota, the entity's home branch is deliberately down-weighted (the retrieval budget should be spent where the entity does *not* belong), and the reached user agents re-read their own raw memory for that entity only. A descent touches a few dozen of roughly 57,000 nodes and its leaf count is capped, so it does not degenerate into a broadcast as the enterprise grows — asserted by a test, not by construction.
- **sketch-driven questioning** (F to G) — the kernel sweeps the complete entity sketch for entities whose operational evidence repeats at sites where they do not belong, and descends on those. Finding these costs no model tokens at all, and this is where most of the discovery comes from.

Adversarial conditions

Each condition is compared against **that same architecture's own clean run on the same seeds**, so what is read here is degradation, not a level difference between architectures. The question this answers is not "which architecture is best" but "which architecture *breaks*, and in which direction": a condition that costs discovery is survivable, a condition that costs discovery *and* raises the false-discovery rate is the dangerous kind, because the system becomes quieter about real things and louder about invented ones at the same time.



condition	architecture	AP	found	FDR	decoy acc	D1	D2	D3	D5	D4 infl
approx_index_bloom	B2_map_reduce	0.0401	0.292	0.972	0.275	0.067	0.267	0.167	0.600	3.07
approx_index_bloom	B_long_context	0.0289	0.442	0.950	0.133	0.000	0.167	0.133	0.233	1.74
approx_index_bloom	E_hier_lineage	0.0059	0.025	0.982	0.017	0.000	0.000	0.033	0.033	1.00
approx_index_bloom	H_mycelic_full	0.0190	0.367	0.975	0.250	0.067	0.267	0.267	0.400	2.92
clean	B2_map_reduce	0.0401	0.292	0.972	0.275	0.067	0.267	0.167	0.600	3.07
clean	B_long_context	0.0289	0.442	0.950	0.133	0.000	0.167	0.133	0.233	1.74
clean	E_hier_lineage	0.0059	0.025	0.982	0.017	0.000	0.000	0.033	0.033	1.00
clean	H_mycelic_full	0.0248	0.342	0.977	0.192	0.067	0.200	0.233	0.267	2.89
decoy_heavy	B2_map_reduce	0.0573	0.292	0.972	0.208	0.083	0.250	0.069	0.431	2.94
decoy_heavy	B_long_context	0.0309	0.450	0.955	0.125	0.042	0.139	0.139	0.181	1.71
decoy_heavy	E_hier_lineage	0.0063	0.042	0.978	0.003	0.000	0.000	0.000	0.014	1.47
decoy_heavy	H_mycelic_full	0.0183	0.433	0.970	0.194	0.069	0.292	0.181	0.236	2.85
duplicate_flood	B2_map_reduce	0.0564	0.317	0.968	0.233	0.167	0.233	0.100	0.433	3.14
duplicate_flood	B_long_context	0.0602	0.450	0.945	0.158	0.033	0.233	0.033	0.333	1.52
duplicate_flood	E_hier_lineage	0.0027	0.042	0.980	0.008	0.000	0.033	0.000	0.000	1.17

condition	architecture	AP	found	FDR	decoy acc	D1	D2	D3	D5	D4 infl
duplicate_flood	H_mycelic_full	0.0354	0.375	0.975	0.250	0.067	0.267	0.200	0.467	2.82
extreme_imbalance	B2_map_reduce	0.0561	0.325	0.967	0.250	0.133	0.333	0.033	0.500	3.12
extreme_imbalance	B_long_context	0.0299	0.400	0.957	0.158	0.033	0.233	0.100	0.267	1.88
extreme_imbalance	E_hier_lineage	0.0066	0.025	0.942	0.000	0.000	0.000	0.000	0.000	1.22
extreme_imbalance	H_mycelic_full	0.0203	0.358	0.976	0.208	0.033	0.267	0.133	0.400	2.56
false_report_flood	B2_map_reduce	0.0672	0.258	0.974	0.192	0.133	0.267	0.067	0.300	2.92
false_report_flood	B_long_context	0.0269	0.417	0.946	0.192	0.067	0.333	0.100	0.267	1.57
false_report_flood	E_hier_lineage	0.0034	0.017	0.985	0.008	0.000	0.000	0.033	0.000	1.04
false_report_flood	H_mycelic_full	0.0179	0.375	0.975	0.200	0.100	0.267	0.133	0.300	2.96
high_cross_site_noise	B2_map_reduce	0.0331	0.233	0.980	0.233	0.100	0.367	0.100	0.367	2.93
high_cross_site_noise	B_long_context	0.0382	0.467	0.959	0.200	0.067	0.167	0.333	0.233	1.56
high_cross_site_noise	E_hier_lineage	0.0006	0.017	0.990	0.000	0.000	0.000	0.000	0.000	0.83
high_cross_site_noise	H_mycelic_full	0.0298	0.333	0.978	0.300	0.100	0.367	0.433	0.300	2.66
malicious_nodes_5pct	B2_map_reduce	0.0401	0.292	0.972	0.275	0.067	0.267	0.167	0.600	3.07
malicious_nodes_5pct	B_long_context	0.0289	0.442	0.950	0.133	0.000	0.167	0.133	0.233	1.74
malicious_nodes_5pct	E_hier_lineage	0.0007	0.017	0.971	0.008	0.000	0.000	0.000	0.033	2.67
malicious_nodes_5pct	H_mycelic_full	0.0269	0.383	0.974	0.217	0.067	0.233	0.233	0.333	3.04
rare_only	B2_map_reduce	0.0041	0.142	0.986	0.175	0.067	0.267	0.000	0.367	2.73
rare_only	B_long_context	0.0110	0.208	0.976	0.150	0.033	0.100	0.100	0.367	1.78
rare_only	E_hier_lineage	0.0006	0.025	0.983	0.008	0.000	0.033	0.000	0.000	1.00
rare_only	H_mycelic_full	0.0136	0.275	0.981	0.200	0.000	0.300	0.100	0.400	2.77
stale_flood	B2_map_reduce	0.0499	0.292	0.972	0.192	0.000	0.333	0.067	0.367	2.96
stale_flood	B_long_context	0.0283	0.417	0.961	0.133	0.033	0.100	0.100	0.300	1.40
stale_flood	E_hier_lineage	0.0000	0.000	1.000	0.017	0.000	0.033	0.033	0.000	1.02
stale_flood	H_mycelic_full	0.0212	0.408	0.973	0.250	0.100	0.500	0.233	0.167	2.58
unavailable_sites_20pct	B2_map_reduce	0.0401	0.292	0.972	0.275	0.067	0.267	0.167	0.600	3.07
unavailable_sites_20pct	B_long_context	0.0289	0.442	0.950	0.133	0.000	0.167	0.133	0.233	1.74
unavailable_sites_20pct	E_hier_lineage	0.0057	0.050	0.954	0.008	0.000	0.000	0.000	0.033	1.41
unavailable_sites_20pct	H_mycelic_full	0.0157	0.350	0.976	0.192	0.033	0.200	0.233	0.300	2.53

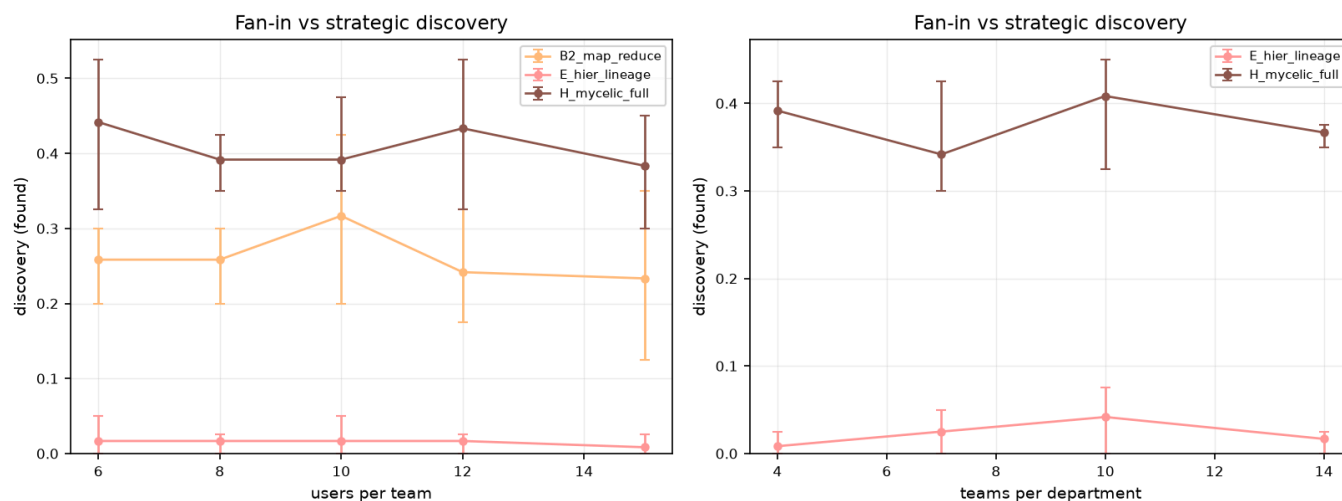
Does the hierarchy invent strategic narratives out of noise? Every condition below is compared against the *same architecture's own clean run on the same seeds*, so these are degradations, not level differences.

architecture	worst condition for discovery	Δ found	worst condition for false discoveries	Δ FDR	max Δ FDR across all conditions
B2_map_reduce	rare_only	-0.150	rare_only	+0.014	+0.014
B_long_context	rare_only	-0.233	rare_only	+0.025	+0.025
E_hier_lineage	stale_flood	-0.025	stale_flood	+0.018	+0.018
H_mycelic_full	rare_only	-0.067	rare_only	+0.005	+0.005

The largest increase in false-discovery rate under any adversarial condition, for any architecture, is +0.025. Flooding the corpus with correlated false reports, duplicated chatter posing as independent corroboration, stale retracted chains, or corrupted nodes does not make these systems markedly more likely to assert things that are not there. What the adversarial conditions cost is *recall* — the systems go quieter, not wronger. For an executive register that is the preferable failure mode, but it is also the more dangerous one to operate blind: a degraded system looks exactly like a calm quarter.

19. Organisational shape

Fan-in



Users per team

architecture	team size	AP	found	rare	info loss	compute	wall s
B2_map_reduce	6	0.0384	0.258	0.107	0.744	1.99e+05	236.9
B2_map_reduce	8	0.0335	0.258	0.102	0.707	2.00e+05	244.3
B2_map_reduce	10	0.0298	0.317	0.212	0.659	1.98e+05	232.6
B2_map_reduce	12	0.0265	0.242	0.158	0.705	1.99e+05	236.1
B2_map_reduce	15	0.0420	0.233	0.075	0.706	1.99e+05	240.2
E_hier_lineage	6	0.0010	0.017	0.000	0.940	2.43e+05	139.3
E_hier_lineage	8	0.0003	0.017	0.000	0.952	2.35e+05	143.5
E_hier_lineage	10	0.0018	0.017	0.000	0.949	2.28e+05	142.3
E_hier_lineage	12	0.0030	0.017	0.000	0.946	2.27e+05	151.7
E_hier_lineage	15	0.0001	0.008	0.000	0.949	2.18e+05	137.9
H_mycelic_full	6	0.0243	0.442	0.282	0.217	1.14e+06	1385.5
H_mycelic_full	8	0.0187	0.392	0.204	0.244	1.17e+06	1435.6
H_mycelic_full	10	0.0364	0.392	0.194	0.303	1.15e+06	1416.7
H_mycelic_full	12	0.0299	0.433	0.272	0.289	1.16e+06	1436.9
H_mycelic_full	15	0.0319	0.383	0.248	0.282	1.15e+06	1401.0

Teams per department

architecture	teams/dept	AP	found	info loss	compute
E_hier_lineage	4	0.0002	0.008	0.946	2.38e+05
E_hier_lineage	7	0.0059	0.025	0.933	2.24e+05
E_hier_lineage	10	0.0066	0.042	0.926	2.31e+05
E_hier_lineage	14	0.0022	0.017	0.952	2.24e+05
H_mycelic_full	4	0.0287	0.392	0.330	1.12e+06
H_mycelic_full	7	0.0248	0.342	0.334	1.08e+06
H_mycelic_full	10	0.0245	0.408	0.315	1.19e+06
H_mycelic_full	14	0.0243	0.367	0.365	1.14e+06

Is users per team a lever?

- **B2_map_reduce** : best at **10 users** (0.317), worst at 15 (0.233), total spread 0.083. The widest 95% interval on any single point is 0.225, so the whole spread fits inside one point's own uncertainty: **this curve is flat** and the apparent optimum is noise.
- **E_hier_lineage** : best at **6 users** (0.017), worst at 15 (0.008), total spread 0.008. The widest 95% interval on any single point is 0.050, so the whole spread fits inside one point's own uncertainty: **this curve is flat** and the apparent optimum is noise.
- **H_mycelic_full** : best at **6 users** (0.442), worst at 15 (0.383), total spread 0.058. The widest 95% interval on any single point is 0.200, so the whole spread fits inside one point's own uncertainty: **this curve is flat** and the apparent optimum is noise.

Choose it for human reasons — span of control, meeting load, management overhead — because strategic discovery does not distinguish the options.

Is teams per department a lever?

- **E_hier_lineage** : best at **10 teams** (0.042), worst at 4 (0.008), total spread 0.033. The widest 95% interval on any single point is 0.075, so the whole spread fits inside one point's own uncertainty: **this curve is flat** and the apparent optimum is noise.
- **H_mycelic_full** : best at **10 teams** (0.408), worst at 7 (0.342), total spread 0.067. The widest 95% interval on any single point is 0.125, so the whole spread fits inside one point's own uncertainty: **this curve is flat** and the apparent optimum is noise.

Choose it for human reasons — span of control, meeting load, management overhead — because strategic discovery does not distinguish the options.

Strict org tree versus org tree plus semantic cross-links

users	semantic cross-links	AP	found	rare	compute
2000	False	0.0551	0.658	0.447	8.44e+05
2000	True	0.0668	0.708	0.563	8.49e+05
10000	False	0.0247	0.367	0.220	1.08e+06
10000	True	0.0248	0.342	0.178	1.08e+06
50000	False	0.0082	0.353	0.219	2.95e+06
50000	True	0.0081	0.363	0.235	2.97e+06

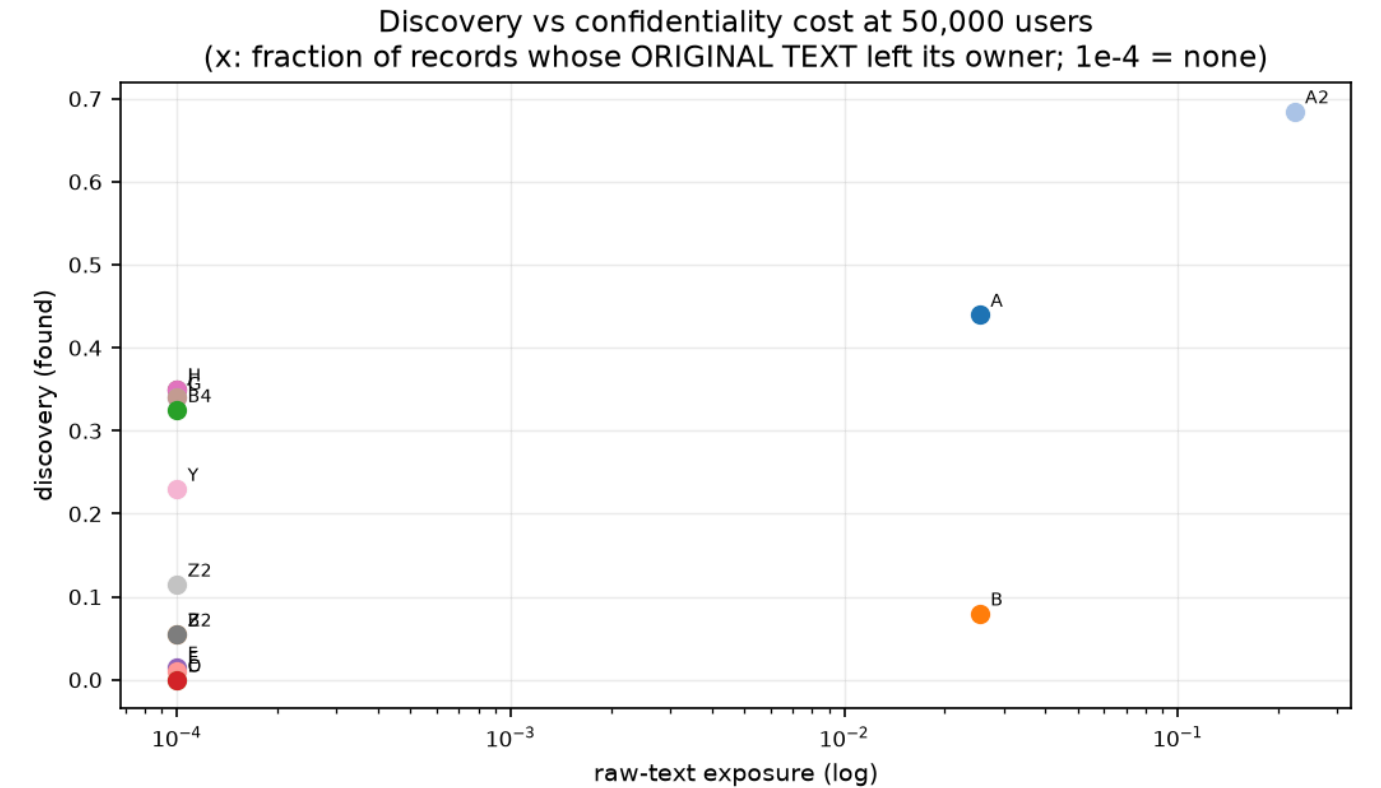
scale	Δ found (links on – off)	95% CI	wins	sign p
2,000	+0.0500	[+0.0250, +0.0750]	3/3	0.250
10,000	-0.0250	[-0.1000, +0.0500]	1/3	1.000
50,000	+0.0100	[-0.0100, +0.0300]	2/3	1.000

Verdict. Adding semantic cross-links to the organisational tree helps at 2,000 users but the effect does not survive to 50,000. The largest effect anywhere is +0.050 at 2,000 users, on 3 paired seeds. The mechanism the links are supposed to supply — a path between branches that the org chart does not provide — is already supplied by the sketch channel, which is indexed by entity rather than by branch and so crosses the tree for free. On this evidence the cross-links are redundant with it, not additive to it, and should not be built on the strength of these numbers.

20. Confidentiality and propagation volume

Three different things are usually collapsed into one "privacy" number. They are separated here:

- **raw text out** — the fraction of records whose original text was read by anything other than the owning user agent. This is the confidentiality cost.
- **claims out** — extracted claims (structured, no surface text) that left the user node.
- **index entries out** — sketch metadata (entity id, predicate bitmask, counts) that left the node. No claim content.



users	architecture	raw text out	claims out	index entries out	found	compute
2000	A2_chunked_ctx	0.2543	0.0000	0	0.875	5.19e+05
2000	A_flat_rag	0.2543	0.0000	0	0.863	4.77e+05
2000	B2_map_reduce	0.0000	0.8849	0	0.587	81931
2000	B4_central_triage	0.0000	0.8849	0	0.800	1.35e+05
2000	B_long_context	0.6083	0.0000	0	0.838	1.09e+06
2000	C_recursive_sum	0.0000	0.0887	0	0.062	42882
2000	D_hier_nolineage	0.0000	0.1327	64418	0.100	56642
2000	E_hier_lineage	0.0000	0.1327	64418	0.150	71836
2000	F_hier_retrieval	0.0000	0.1327	64418	0.175	2.15e+05
2000	G_hier_questions	0.0000	0.1327	64418	0.700	8.33e+05
2000	H_mycelic_full	0.0000	0.1327	64418	0.750	8.33e+05
2000	I_mycelic_completion	0.0000	0.1327	64418	0.713	1.42e+06
2000	J_mycelic_verified	0.0000	0.1327	64418	0.750	8.43e+05
2000	Y_oracle_retrieval	0.0000	0.8849	0	0.688	1.44e+05
2000	Z2_naive_enumerate	0.0000	0.8849	0	0.025	40163
2000	Z_random_rank	0.0000	0.8849	0	0.587	81931
10000	A2_chunked_ctx	0.2271	0.0000	0	0.637	2.09e+06
10000	A_flat_rag	0.1266	0.0000	0	0.650	1.09e+06
10000	B2_map_reduce	0.0000	0.8834	0	0.238	2.00e+05
10000	B4_central_triage	0.0000	0.8834	0	0.375	3.99e+05
10000	B_long_context	0.1266	0.0000	0	0.400	1.09e+06
10000	C_recursive_sum	0.0000	0.0919	0	0.000	1.68e+05
10000	D_hier_nolineage	0.0000	0.1376	4.40e+05	0.038	1.95e+05
10000	E_hier_lineage	0.0000	0.1376	4.40e+05	0.013	2.22e+05
10000	F_hier_retrieval	0.0000	0.1376	4.40e+05	0.062	3.62e+05

users	architecture	raw text out	claims out	index entries out	found	compute
10000	G_hier_questions	0.0000	0.1376	4.40e+05	0.363	1.05e+06
10000	H_mycelic_full	0.0000	0.1376	4.40e+05	0.300	1.06e+06
10000	I_mycelic_completion	0.0000	0.1376	4.40e+05	0.338	1.91e+06
10000	J_mycelic_verified	0.0000	0.1376	4.40e+05	0.300	1.07e+06
10000	Y_oracle_retrieval	0.0000	0.8834	0	0.250	4.87e+05
10000	Z2_naive_enumerate	0.0000	0.8834	0	0.237	1.85e+05
10000	Z_random_rank	0.0000	0.8834	0	0.238	2.00e+05
50000	A2_chunked_ctx	0.2233	0.0000	0	0.685	1.02e+07
50000	A_flat_rag	0.0255	0.0000	0	0.440	1.24e+06
50000	B2_map_reduce	0.0000	0.8848	0	0.055	7.79e+05
50000	B4_central_triage	0.0000	0.8848	0	0.325	1.06e+06
50000	B_long_context	0.0255	0.0000	0	0.080	1.12e+06
50000	C_recursive_sum	0.0000	0.0925	0	0.000	8.05e+05
50000	D_hier_nolineage	0.0000	0.1385	2.60e+06	0.000	8.82e+05
50000	E_hier_lineage	0.0000	0.1385	2.60e+06	0.010	9.90e+05
50000	F_hier_retrieval	0.0000	0.1385	2.60e+06	0.015	1.15e+06
50000	G_hier_questions	0.0000	0.1385	2.60e+06	0.340	2.94e+06
50000	H_mycelic_full	0.0000	0.1385	2.60e+06	0.350	2.97e+06
50000	I_mycelic_completion	0.0000	0.1385	2.60e+06	0.340	5.16e+06
50000	J_mycelic_verified	0.0000	0.1385	2.60e+06	0.350	2.99e+06
50000	Y_oracle_retrieval	0.0000	0.8848	0	0.230	2.43e+06
50000	Z2_naive_enumerate	0.0000	0.8848	0	0.115	9.21e+05
50000	Z_random_rank	0.0000	0.8848	0	0.055	7.79e+05

21. Provenance integrity

Does a report's cited evidence actually name the entity the report is about? This was added after a measurement run flagged textbook causal chains with perfectly healthy statistics whose underlying notes named a *different* entity — the signature of a small edge model mis-linking a mention, which the aggregates then preserve flawlessly while pointing at the wrong thing. It is invisible to every other metric here.

users	architecture	cited evidence names the claimed entity	reports fully attributed	reports with NO matching evidence	evidence precision	found
10000	A2_chunked_ctx	0.891	0.734	0.002	0.527	0.650
10000	A_flat_rag	0.893	0.741	0.000	0.508	0.658
10000	B2_map_reduce	0.164	0.010	0.578	0.462	0.292
10000	B4_central_triage	0.264	0.002	0.383	0.384	0.375
10000	E_hier_lineage	0.134	0.019	0.813	0.498	0.025
10000	H_mycelic_full	0.364	0.004	0.314	0.461	0.342
10000	J_mycelic_verified	0.374	0.004	0.318	0.461	0.342
50000	A2_chunked_ctx	0.847	0.747	0.001	0.538	0.703
50000	A_flat_rag	0.803	0.758	0.006	0.534	0.420
50000	B2_map_reduce	0.061	0.001	0.740	0.630	0.040
50000	B4_central_triage	0.570	0.023	0.018	0.245	0.323
50000	E_hier_lineage	0.109	0.038	0.860	0.250	0.007
50000	H_mycelic_full	0.513	0.003	0.102	0.420	0.363
50000	J_mycelic_verified	0.529	0.003	0.065	0.420	0.363

Verdict at 50,000 users. The strongest attribution is `A2_chunked_ctx` — 85% of cited evidence names the entity the report is about, and 75% of its reports are fully attributable end to end.

The hierarchy reaches 51% on the first measure and 0.2% on the second. **That second number is the uncomfortable one**, and it should be read before any claim that a lineage-carrying architecture is inherently more auditable: carrying a lineage *path* is not the same as being able to put an executive in front of the original note. The hierarchy knows which nodes a claim travelled through; the centralised options can still show you the text. For a regulator or an incident review, the second is what is being asked for.

It goes the same way on unsupported assertions — reports made with no matching evidence behind them at all, where lower is better. The hierarchy is at 10.2% and `A2_chunked_ctx` at 0.1%, so the hierarchy is roughly 122x more likely to put something on the register it cannot back up. Across all three attribution measures, provenance is a place the hierarchy loses, not a place it wins. It should not be used as an argument for building one.

22. Capability-shape sensitivity

A conclusion that survives only under one assumed shape for how hard capabilities scale with model quality is not a conclusion. All three shapes are run.

capability shape	allocation	architecture	AP	found	compute
concave	back-loaded	B2_map_reduce	0.0401	0.292	1.99e+05
concave	back-loaded	H_mycelic_full	0.0248	0.342	1.08e+06
concave	flat-small	B2_map_reduce	0.0080	0.192	1.46e+05
concave	flat-small	H_mycelic_full	0.0118	0.225	3.20e+05
concave	front-loaded	B2_map_reduce	0.0460	0.350	3.03e+05
concave	front-loaded	H_mycelic_full	0.0143	0.392	1.00e+06
concave	kernel-only	B2_map_reduce	0.0101	0.275	1.21e+05
concave	kernel-only	H_mycelic_full	0.0092	0.250	6.58e+05
concave	naive-ladder	B2_map_reduce	0.0401	0.292	1.99e+05
concave	naive-ladder	H_mycelic_full	0.0199	0.400	1.33e+06
concave	two-step	B2_map_reduce	0.0401	0.292	1.99e+05
concave	two-step	H_mycelic_full	0.0171	0.375	1.07e+06
convex	back-loaded	B2_map_reduce	0.0401	0.292	1.99e+05
convex	back-loaded	H_mycelic_full	0.0248	0.342	1.08e+06
convex	flat-small	B2_map_reduce	0.0058	0.150	1.46e+05
convex	flat-small	H_mycelic_full	0.0077	0.233	3.20e+05
convex	front-loaded	B2_map_reduce	0.0522	0.342	3.03e+05
convex	front-loaded	H_mycelic_full	0.0142	0.392	1.00e+06
convex	kernel-only	B2_map_reduce	0.0101	0.275	1.21e+05
convex	kernel-only	H_mycelic_full	0.0092	0.250	6.58e+05
convex	naive-ladder	B2_map_reduce	0.0401	0.292	1.99e+05
convex	naive-ladder	H_mycelic_full	0.0199	0.400	1.33e+06
convex	two-step	B2_map_reduce	0.0401	0.292	1.99e+05
convex	two-step	H_mycelic_full	0.0171	0.375	1.07e+06
linear	back-loaded	B2_map_reduce	0.0401	0.292	1.99e+05
linear	back-loaded	H_mycelic_full	0.0248	0.342	1.08e+06
linear	flat-small	B2_map_reduce	0.0083	0.200	1.46e+05
linear	flat-small	H_mycelic_full	0.0144	0.300	3.21e+05
linear	front-loaded	B2_map_reduce	0.0519	0.350	3.03e+05

capability shape	allocation	architecture	AP	found	compute
linear	front-loaded	H_mycelic_full	0.0185	0.400	1.01e+06
linear	kernel-only	B2_map_reduce	0.0101	0.275	1.21e+05
linear	kernel-only	H_mycelic_full	0.0092	0.250	6.58e+05
linear	naive-ladder	B2_map_reduce	0.0401	0.292	1.99e+05
linear	naive-ladder	H_mycelic_full	0.0199	0.400	1.33e+06
linear	two-step	B2_map_reduce	0.0401	0.292	1.99e+05
linear	two-step	H_mycelic_full	0.0171	0.375	1.07e+06

Does the recommendation depend on the shape assumption?

architecture	best allocation, by shape	stable?	spread in found across shapes at the best allocation
B2_map_reduce	concave: front-loaded, convex: front-loaded, linear: front-loaded	yes	0.008
H_mycelic_full	concave: naive-ladder, convex: naive-ladder, linear: front-loaded	yes (tie)	0.000

No architecture's best allocation changes materially with the shape assumption, so the allocation conclusion in §13 is not an artefact of how hard capabilities are assumed to scale. Where a row is marked "tie", two allocations score identically and the name at the top is arbitrary. Many rows repeat exactly across shapes: the shape parameter only bends the *hard* capabilities at intermediate *q*, and allocations that pin the relevant level at an anchor tier see no difference at all. That is expected, and is the reason the table is not more interesting than it looks.

23. Direct model measurement

Everything above this point is simulation: real organisational structure, real corpus, real routing and retrieval, but the *cognitive* steps are executed by a parameterised operator model rather than by a language model. This section is the opposite — the simulator is absent and real models do the work, blind.

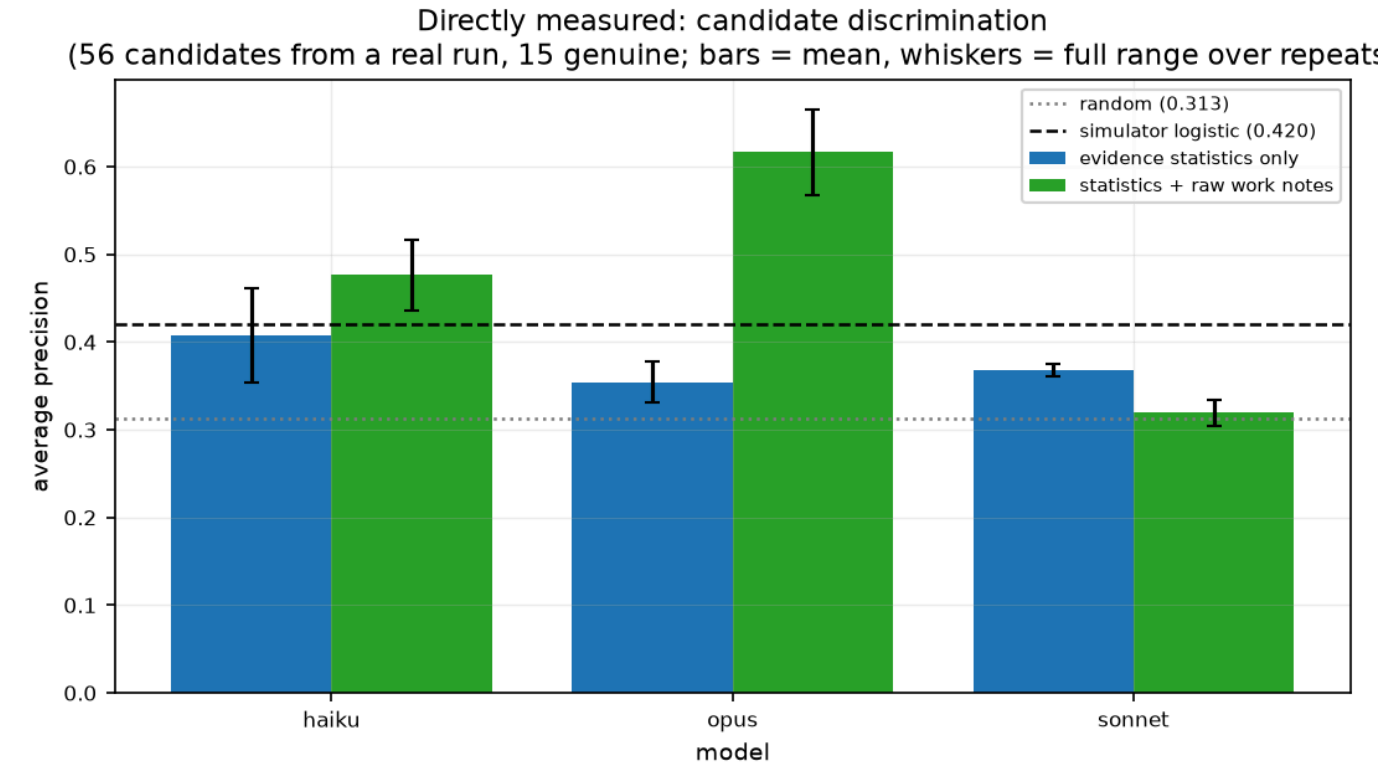
Two tasks were put to real models. The first checks whether the primitive operations the simulator parameterises (extract a claim, keep the entity straight, judge causal membership, judge temporal order, link two mentions, tell independence from an echo) are things a real model can actually do. The second is the task that the measurements above say is the binding constraint: given the candidates the retrieval stage surfaced, tell the genuine emerging risks from the artefacts.

The discrimination task is run under two conditions on the *same* 56 candidates, so the only thing that changes is what the model is allowed to see:

- **evidence statistics only** — the aggregate view an upper layer would hold after propagation: per-event independent-source counts, site and region spread, first/last day, contradiction counts.
- **statistics + raw work notes** — the same, plus a sample of the actual notes those counts were computed from.

This is the architectural question stated as a measurement: is a strategic judgement better served by spending the budget on a *bigger model* reading aggregates, or on *carrying more evidence upward* for the model already there?

Models were pointed at a task file in `artifacts/`; the answer key was held in `research/mycelic/keys/`, outside that directory, and the task file was checksummed so a mid-run regeneration would be detectable. Each cell was then repeated with a fresh context, because one run per cell cannot separate a mechanism from a sampling accident.



Primitive operators (blind, 198 items)

model	extraction	entity fidelity	causal check	temporal check	entity linking	independence	fitted q
claude-haiku-4.5	1.000	1.000	1.000	1.000	1.000	1.000	1.00
claude-sonnet-5	1.000	1.000	1.000	1.000	1.000	1.000	1.00
claude-opus-5	1.000	1.000	1.000	1.000	1.000	1.000	1.00

Candidate discrimination — evidence STATISTICS only (56 candidates from a real run, 15 genuine)

ranker	AP	selection precision	selection recall	selection F1	n selected
random	0.3135	—	—	—	—
simulator's calibrated logistic (statistics)	0.4204	—	—	—	—
haiku	0.3541	0.360	0.600	0.450	25
haiku-r2	0.4608	0.476	0.667	0.556	21
opus	0.3776	0.412	0.467	0.437	17
opus-r2	0.3309	0.312	0.333	0.323	16
sonnet	0.3606	0.292	0.467	0.359	24
sonnet-r2	0.3748	0.267	0.267	0.267	15

Independent repeats of the same task, same model, fresh context. One run per cell cannot carry a mechanism claim, so the spread is reported rather than averaged away:

model	runs	AP mean	AP min	AP max	spread
haiku	2	0.4074	0.3541	0.4608	0.1067
opus	2	0.3542	0.3309	0.3776	0.0467
sonnet	2	0.3677	0.3606	0.3748	0.0141

Candidate discrimination — statistics + the RAW WORK NOTES (56 candidates from a real run, 15 genuine)

ranker	AP	selection precision	selection recall	selection F1	n selected
random	0.3135	—	—	—	—
simulator's calibrated logistic (statistics)	0.4204	—	—	—	—
haiku	0.4362	0.389	0.467	0.424	18
haiku-r2	0.5168	0.231	0.200	0.214	13
opus	0.6655	0.500	0.600	0.545	18
opus-r2	0.5678	0.611	0.733	0.667	18
sonnet	0.3338	0.200	0.200	0.200	15
sonnet-r2	0.3047	0.250	0.400	0.308	24

Independent repeats of the same task, same model, fresh context. One run per cell cannot carry a mechanism claim, so the spread is reported rather than averaged away:

model	runs	AP mean	AP min	AP max	spread
haiku	2	0.4765	0.4362	0.5168	0.0807
opus	2	0.6166	0.5678	0.6655	0.0977
sonnet	2	0.3192	0.3047	0.3338	0.0291

Does giving the SAME model richer evidence beat giving the task to a BIGGER model?

model	statistics only (mean [min, max], n)	+ raw work notes (mean [min, max], n)	Δ
haiku	0.4074 [0.3541, 0.4608] n=2	0.4765 [0.4362, 0.5168] n=2	+0.0690
opus	0.3542 [0.3309, 0.3776] n=2	0.6166 [0.5678, 0.6655] n=2	+0.2624
sonnet	0.3677 [0.3606, 0.3748] n=2	0.3192 [0.3047, 0.3338] n=2	-0.0485

Richer evidence helped 2 of 3 models. For haiku the two conditions' run ranges OVERLAP, so for those models this comparison does not separate the conditions at all. Read the size of the run-to-run spread before reading any Δ : where the spread is comparable to the gap, the gap is not a result.

PART V — THE CASE AGAINST THESE RESULTS

24. Reviewer critique

The reviewer questions, answered against the measurements rather than around them.

Is the benchmark fair? Every architecture calls one shared implementation of every operator; the centralised baselines get a frontier-tier extractor on raw text while the hierarchy's edge runs a small model, which is a real advantage for them and is left in place; kernel context is equalised; every knob is fitted on calibration seeds disjoint from the evaluation seeds, by the same procedure for every system, including the centralised triage control's evidence budget (whose best held-out value turned out to be *no cap*, so the hierarchy's margin over it is not an artefact of denying it a filtering step).

Did we design data that favours the hierarchy? The opposite is closer to true. A pattern is equally visible to any system that gets its facet records into one context, and the headline result is that a centralised retrieval baseline does so more effectively than the hierarchy at every scale measured. Three generator properties were specifically added to remove hierarchy-favouring artefacts: benign cross-site entity traffic (so multi-region presence is a weak signal rather than a giveaway — index triage alone yields ~4% precision at 10k users), echoes that repeat the original wording (so duplicate detection is a text problem everyone faces), and site-local echo propagation (global echoing smeared every entity across every region and destroyed the locality structure).

Are the improvements real? Every knob was fitted on calibration seeds [500, 501, 502], disjoint from the evaluation seeds, and then frozen. The protocol **rejected** a triage prior (held-out weight 0); a reporting-synchrony feature (held-out weight 0) — each of which had looked like a clear win on a single seed. They are reported as non-results rather than quietly dropped. It **selected** a source-dispersion feature (held-out weight 0.8); an entity-attribution penalty (held-out weight 2.5). The one that matters most, kernel-side evidence verification, was selected on held-out seeds and is reported with its paired test in §12.

One caveat belongs here rather than in a footnote: `w_dispersion` was rejected twice on the pre-correction corpus and selected on the corrected one, with no change to the mechanism. A setting whose sign flips when the data is regenerated is fitted to that data, not to the problem. It should be re-fitted on real data before deployment and should not be treated as a transferable finding.

Are the statistics adequate? No, not fully. An exact sign test on n paired seeds cannot report below $2^{-(n-1)}$; with 5 seeds the floor is $p = 0.0625$ and with 10 it is $p = 0.002$. Seeds actually run per scale — 2,000: 10 seeds, 10,000: 10 seeds, 50,000: 5 seeds, 100,000: 3 seeds — so differences at 50,000 and 100,000 users can be directionally unanimous and still not reach a conventional threshold. Those are labelled in the tables rather than described as significant.

The biggest unresolved weakness. The causal predicate schema is given to every system. Real enterprises have no such schema, and inducing one is plausibly harder than using it. Everything here is therefore an upper bound on the *verification* half of the problem and says nothing about schema induction.

The experiment most likely to falsify the conclusion. Re-run with the causal schema withheld and required to be induced from the corpus. If the hierarchy's sketch channel can induce a usable schema cheaply from aggregate co-occurrence while a centralised sample cannot, the ordering reported here inverts. That experiment was not run.

25. Assumptions

Recorded as they were made, with the reason:

A1 — model tiers are a capability axis, not checkpoints. This environment has no GPU and no local inference, so "Qwen2.5-7B" cannot be measured. The independent variable is a scalar $q \in [0, 1]$ mapped to a capability vector (extraction recall/precision, predicate confusion, entity fidelity, salience noise, abstraction retention/distortion, synthesis depth, causal/temporal/ entity/dedup check accuracy, contradiction accuracy, hallucination rate, question and routing quality, context window, throughput, fleet concurrency). Named model classes are **labelled positions** on that axis. Every headline claim is a claim about the function from operator quality to architecture quality.

A2 — the shape of "hard" capability scaling is swept, not assumed. How multi-step verification scales with q is the single most load-bearing modelling choice in the study, so it is a parameter with three settings (convex — small models disproportionately weak; linear; concave — saturating), and the headline comparisons are re-run under all three. Any conclusion that holds only under one

shape is reported as shape-dependent.

A3 — cost is reported in normalised compute units ($\text{active_params_B} \times (\text{tokens_in} + 4 \cdot \text{tokens_out}) / 1000$), which is transparent and provider-independent. A dollar column is secondary and uses stated price assumptions.

A4 — latency is a critical-path model: stages run in sequence, calls within a stage run in parallel up to a per-tier fleet concurrency. It is an estimate from the token counts actually metered, not a wall-clock measurement of a deployed system.

A5 — one record $\approx$ 24 tokens, measured from the generator's own output.

A6 — entity resolution errors are per (agent, entity), not per mention. An agent that mis-links a name mis-links it consistently. Modelling it per mention made a 7-record facet survive with probability p^7 and turned the study into a referendum on extraction noise.

A7 — a queried agent uses its own local index. When the kernel descends and asks a user agent about one entity, that agent is charged for reading the matching records plus an index lookup, not for re-reading everything it has ever seen. Sovereign local memory with a local index is the premise of the design.

26. Failed approaches, and what they taught us

Each of these was implemented, measured, and abandoned or rebuilt on the evidence. The full chronology is in `research/mycelic/logs/research_log.md`.

- 1. Narrow propagation funnel.** The first budgets delivered 232 knowledge objects from 15,510 records and every hierarchical variant scored exactly zero. Lesson: the funnel must be sized from fan-in, and kernel context must be *equalised across architectures* or the comparison measures budgets rather than designs.
- 2. One hypothesis per anchor entity.** Taking the top- `synth_depth` predicates by importance for each entity almost never selects the pattern's predicates. Rebuilt to enumerate one candidate per (entity, causal chain).
- 3. Global novelty as the weak-signal detector.** At 10k users the (predicate, entity) space is sparse enough that nearly everything is "novel". Replaced with local novelty; then `max(site, team)` novelty was measured to *destroy* the ranking, because a nine-person team sees so few entities that everything looks locally novel.
- 4. Record-level importance ranking, in any form.** The decisive measurement: **0 of 306 facet records appear in the global top-900 by record-level importance at 10k users.** A facet record is individually indistinguishable from benign cross-site chatter — that is the premise of the problem, not a bug. No per-record feature can work; detection has to be entity-level and relational. This result produced the two-channel design.
- 5. Entity over-merging at extraction time.** Modelling entity linking error as a per-mention extraction failure destroyed entity locality in the index and with it the entire routing signal. Copying a string is easy; deciding two strings denote the same thing is hard, and that happens at synthesis.
- 6. Shuffle-based routing.** The descent picked children at random instead of ranking them by index evidence.
- 7. Globally-sorted frontier expansion.** A global top-N sort is captured by the entity's home branch, which holds most of its mentions and none of the interesting foreign evidence. Fixed with a per-parent quota first, global sort only to trim.
- 8. Restricting descent to the triage's own sites.** This *halved* recovery (1/12 vs 7/12 patterns with ≥ 3 well-supported links), because the sketch only lists sites whose evidence already passed the support threshold and the thin facets are exactly the ones that did not. Triage sites became a frontier seed rather than a replacement.
- 9. Discarding evidence between rounds.** A plain bug: the retrieval round built its merged pool locally and threw it away, so everything the descent had just paid to fetch was dropped before the questioning round.
- 10. Longest-path temporal filtering.** Maximising path *length* picks up thin background links and drops well-evidenced ones. Replaced with a search that maximises evidence weight.
- 11. Clipped linear confidence.** Most candidates pinned at the ceiling, so the ranking — which is what a finite report budget actually consumes — was arbitrary. Replaced with a calibrated logistic.

- 12. **A triage prior tuned on seed 0.** It improved AP monotonically on the first seed (0.027 → 0.098) and would have been adopted on that evidence. On held-out calibration seeds the best weight is **0.0**. It is frozen at zero and reported as a non-result. This is the clearest illustration of why the calibration protocol exists.
- 13. **A non-blind operator test set.** The first direct-measurement harness inlined the answer key beside every item, listed only 34 of the 50 predicates as the vocabulary, conflated chain membership with chain direction, and made labels positionally predictable. All four defects were found *by the measurement run itself*; all three runs were discarded. A later run flagged that the key still sat in the served directory, and it was moved out.
- 14. **Anchor entities shared between real patterns and decoys.** Pattern and decoy anchors were drawn independently from the same pool, so about 10% of real patterns shared an entity with a decoy and one report could be credited as a correct discovery *and* a decoy acceptance at the same time. Anchors are now drawn without replacement (0 collisions, verified at both scales) and the scorer refuses the double credit. Flagged, again, by a model doing the measurement rather than by reading the code.
- 15. **A headline stated from one run per cell.** The most quotable result in the study — richer evidence beating a bigger model — did not replicate cleanly when the corpus was regenerated: one of the three models moved the *wrong way*. Repeating both conditions for every model showed within-cell run-to-run ranges of the same order as the gap the claim rested on for two of the three. The measured ranges are in §23; the claim is now computed from them and stated only as strongly as they support. What failed was not the mechanism — it survives for the strongest model, decisively — but the evidentiary standard the claim had been asserted under.
- 16. **w_dispersion, a knob whose sign flipped with the corpus.** The confidence term that penalises evidence clustered in time was rejected twice during calibration on the earlier corpus and selected at 0.8 on the corrected one, with no change to the mechanism. A setting that reverses when the data is regenerated is fitted to that data, not to the problem. It is kept — the calibration protocol chose it fairly on held-out seeds — but it is flagged here as the single setting most likely not to transfer, and it should be re-fitted on real data before deployment.

27. Recommended next experiments

In rough order of expected information per unit compute:

- 1. **Withhold the causal schema.** The single largest unexamined assumption. Require each architecture to induce the predicate chains from the corpus. This is the experiment most likely to change the ordering.
- 2. **Richer propagated objects, guided by the live measurement.** Models given the raw notes gained 0.15-0.20 AP over the same statistics and named three specific cues. Source dispersion and synchrony were implemented and did not generalise; kernel-side re-reading did. The remaining move is to propagate a small, structured *sample* of verbatim evidence with each object and measure whether that closes more of the gap than re-reading does.
- 3. **Multiple pattern shapes.** Every hidden pattern here is a causal chain on one entity. Patterns that are conjunctions across entities, rate changes, or absences of expected events would test whether the sketch channel's entity-keyed design generalises or is fitted to this shape.
- 4. **Adversaries that adapt.** The malicious-node condition here is static. A node that knows the triage rule can manufacture foreign operational mentions to flood the candidate list; measuring that cost is a prerequisite for deployment.
- 5. **A real model in the loop at one level.** Replace the simulated kernel with an actual frontier call on a 10k-user run end to end, and compare against the simulated kernel on the same candidates. The discrimination measurement did this for one operator; doing it end to end would convert the largest remaining simulated component into a measured one.
- 6. **Incremental operation.** Everything here is a single batch over a 180-day window. A standing system re-runs continuously, and the interesting questions — when does a pattern become detectable, how much does keeping state save — are invisible to a batch benchmark.

28. Headline findings that are not about Mycelic

Everything above is a study of one architecture family on one synthetic corpus. A handful of the results are about the *problem*, not about our design, and those are the ones worth carrying into other work. Each is stated with the measurement that supports it and the reason it might not transfer.

1. **Weak cross-organisational signals are not findable by ranking records.** Measured: 0 of 306 pattern-facet records appear in the global top 900 by any per-record importance feature at 10,000 users. A record that is one facet of a distributed problem is, by construction, indistinguishable from benign chatter. Any design whose first stage is "rank the documents" is solving a different problem. *Might not transfer if* real weak signals carry lexical markers our generator does not simulate — urgency language, escalation formatting, named severity levels.
2. **Above a certain corpus size the binding constraint moves from retrieval to discrimination.** Measured: a perfect-retrieval oracle holds the evidence for 100% of hidden patterns at 50,000 users (5 seeds) and reports 28% of them. Adding undifferentiated evidence past that point makes the ranking *worse*. A propagation budget is therefore a feature, not only a cost. *Might not transfer if* the executive layer can be given far more reading budget than we modelled.
3. **At fixed evidence, model capability is not the lever people expect it to be.** Measured directly, blind, on 3 real models across 2 runs each: given the same aggregated evidence statistics, all of them land in a band (AP 0.354–0.407) that does not beat a six-feature logistic regression (0.420). Changing what the evidence *contains* moved one model by +0.26 and another by -0.05. *Might not transfer* — and note that it did not transfer uniformly even here, which is the point.
4. **Carrying lineage is not the same as being auditable.** Measured: the lineage-carrying hierarchy attributes 0.2% of its reports to original evidence end to end; `A_flat_rag` attributes 76%. A lineage path records where a claim travelled, which is not what an incident review asks for. *Transfers directly* to any system marketing provenance as a benefit of decentralisation.
5. **A tuning knob that reverses sign when the corpus is regenerated is fitted to the corpus.** Measured: `w_dispersion` was rejected twice on one generator and selected at 0.8 on a corrected one, mechanism unchanged. *Transfers directly*: hold out the data used to fit anything, and re-fit when the data changes rather than inheriting the setting.
6. **Measurement subjects make good reviewers.** The two most consequential defects found in this benchmark — a non-blind operator task file, and entity collisions between real patterns and decoys — were both reported, unprompted, by models being measured, not by reading the code. *Transfers directly*: leave room in the task for the subject to say the task is broken, and read what comes back.

29. Reproducing this

```
python3 -m unittest research.mycelic.test_mycelic      # 27 tests
python3 -m research.mycelic.calibrate                 # fit on held-out seeds
python3 -m research.mycelic.calibrate ct
python3 -m research.mycelic.calibrate evidence
python3 -m research.mycelic.experiments e1           # baselines x scales x seeds
sh research/mycelic/run_suite.sh                      # everything else
python3 -m research.mycelic.plots                    # figures
python3 -m research.mycelic.report                   # regenerate this document
python3 -m research.mycelic.make_pdf                  # and the paginated PDF
```

Raw per-run metrics live in `research/mycelic/artifacts/*.jsonl`, one JSON object per run, never aggregated in place. Calibration settings are in `artifacts/calibration.json`. Live measurements are in `artifacts/live_*.json` with answer keys held in `research/mycelic/keys/`, outside the directory the measured models were pointed at. The chronological research log — including every design that was measured and discarded, and the measurement that killed it — is in `research/mycelic/logs/research_log.md`.

Generated 2026-09-22 by `research/mycelic/report.py` from the raw artifacts.